

# Electric Circuits

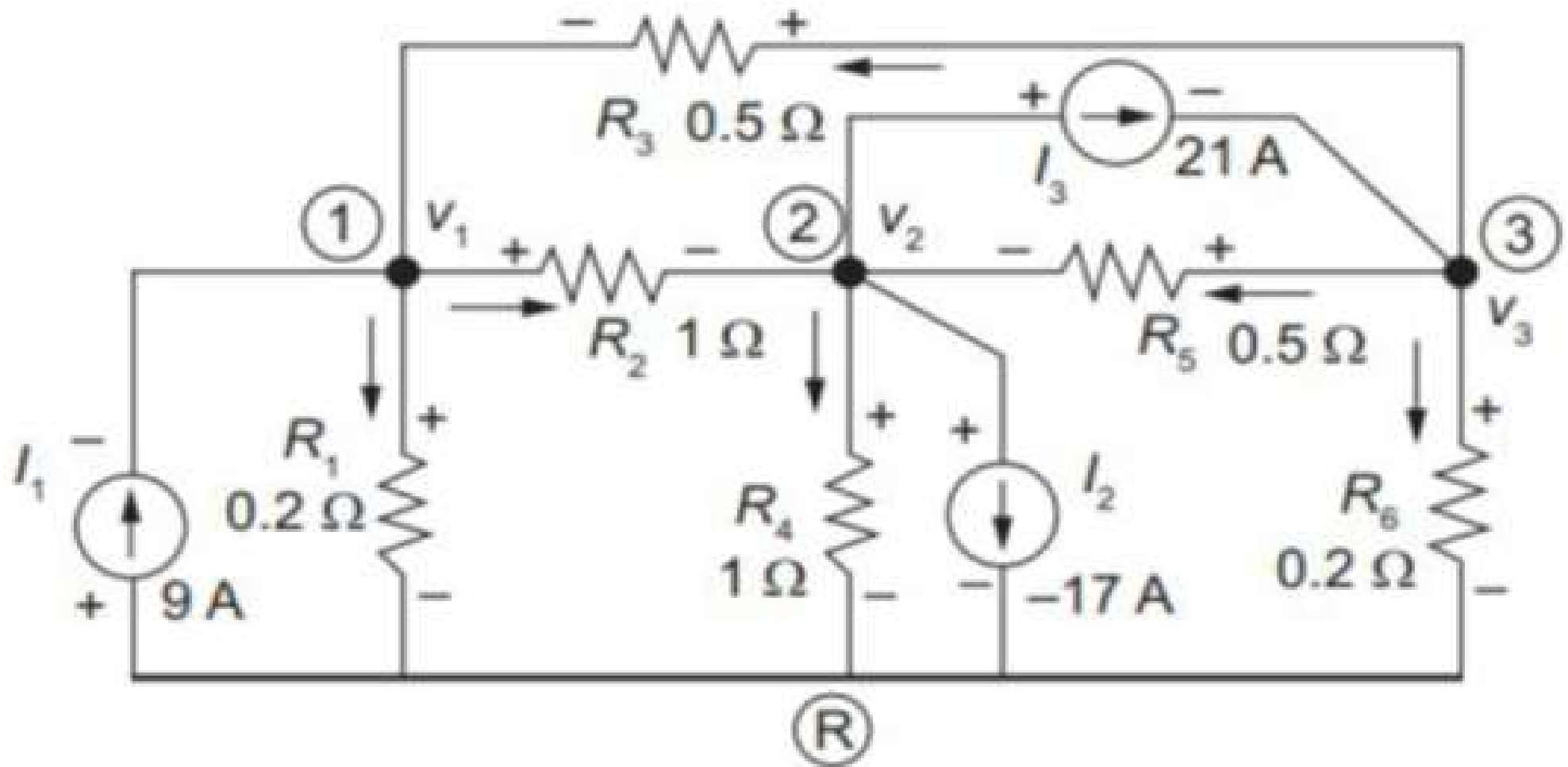


*Network Theorems:  
Superposition Theorem*

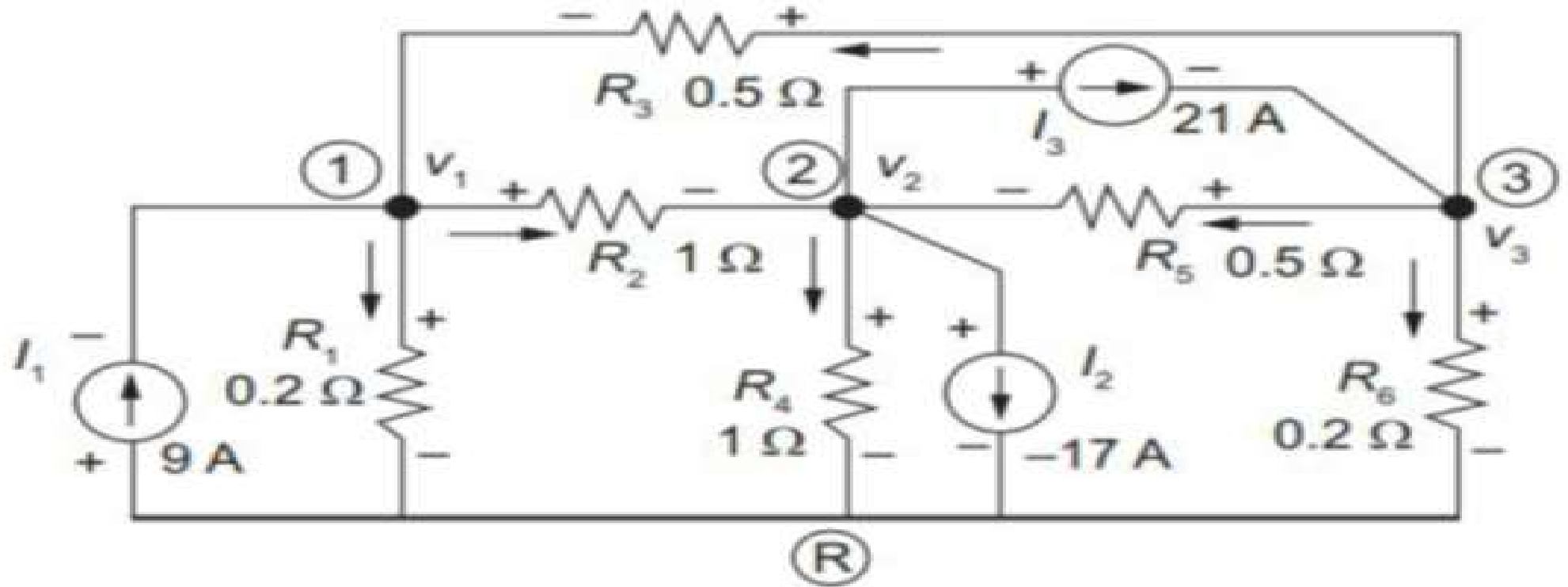
# Tentative Lecture Plan

| L                                     | LDay-1      | LDay-2      | Topics                                                                              | T | TDay        |
|---------------------------------------|-------------|-------------|-------------------------------------------------------------------------------------|---|-------------|
| 1                                     | FRI, 04 AUG | MON, 07 AUG | Basic Definitions, Voltage & Current Sources, KCL & KVL                             | 1 | THU, 10 AUG |
| 2                                     | TUE, 08 AUG | FRI, 11 AUG | Nodal Analysis: Independent and Dependent Current Sources                           | 2 | SAT, 19 AUG |
| 3                                     | MON, 14 AUG | THU, 17 AUG | Nodal Analysis: Current and Voltage Sources                                         |   |             |
| 4                                     | FRI, 18 AUG | MON, 21 AUG | Mesh Analysis: Voltage and Current Sources                                          | 3 | THU, 24 AUG |
| 5                                     | TUE, 22 AUG | FRI, 25 AUG | Network Theorems: Superposition Theorem                                             |   |             |
| SUN, 27 AUG : Quiz-1 : Lectures 1-5   |             |             |                                                                                     |   |             |
| 6                                     | FRI, 26 AUG | MON, 28 AUG | Network Theorems: Thevenin's and Norton's Theorem                                   | 4 | THU, 31 AUG |
| 7                                     | TUE, 29 AUG | FRI, 04 SEP | Network Theorems: Maximum Power Transfer Theorem                                    | 5 | WED, 06 SEP |
| 8                                     | TUE, 05 SEP | FRI, 08 SEP | Series RL and RC Circuits                                                           |   |             |
| 9                                     | SAT, 09 SEP | MON, 11 SEP | Series RL, RC, RLC Circuits<br>Response to Constant, Sinusoid and Exponential Input | 6 | THU, 14 SEP |
| 10                                    | TUE, 12 SEP | FRI, 15 SEP | Concept of Phasor, AC Circuits, AC Power                                            | R | SAT, 16 SEP |
| WED, 20 SEP: MidSem : Lectures 1 – 10 |             |             |                                                                                     |   |             |

## Solution by Nodal Analysis

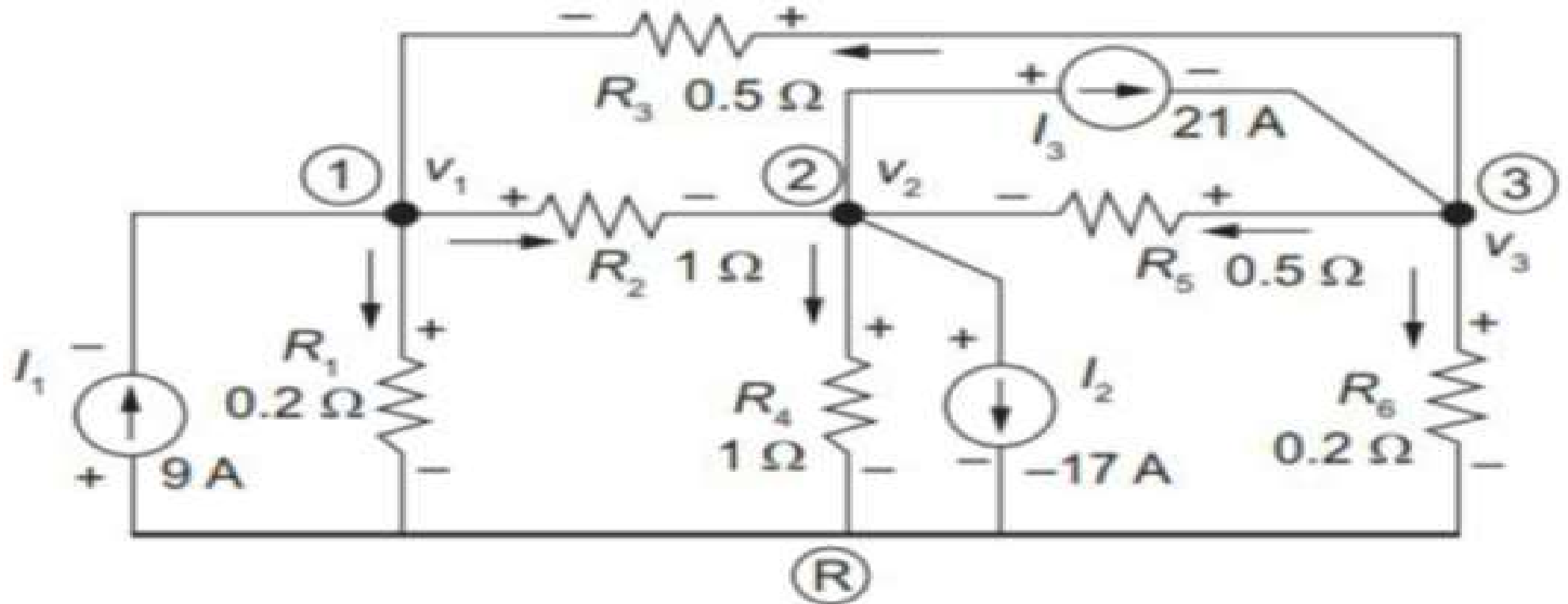


## Solution by Nodal Analysis



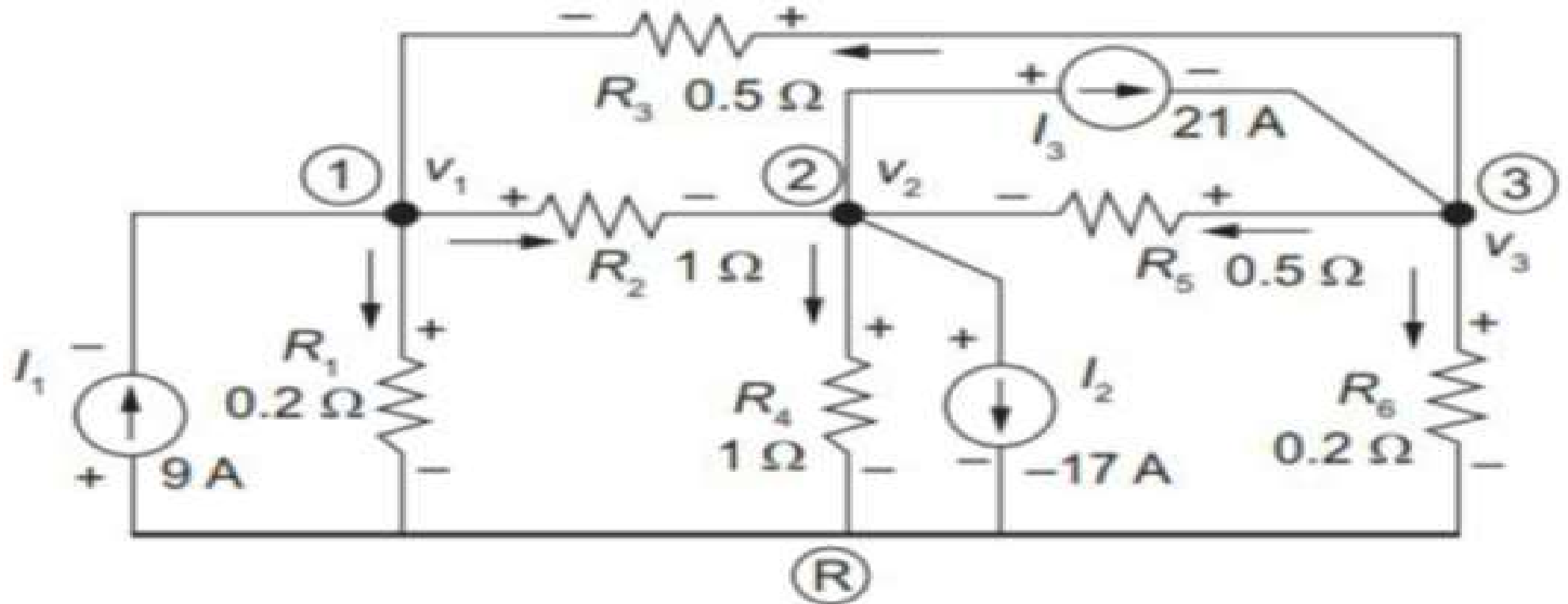
$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}$$

## Solution by Nodal Analysis



$$\begin{bmatrix} 8 & -1 & -2 \\ -1 & 4 & -2 \\ -2 & -2 & 9 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 9 \\ -17 \\ 21 \end{bmatrix}$$

## Solution by Nodal Analysis

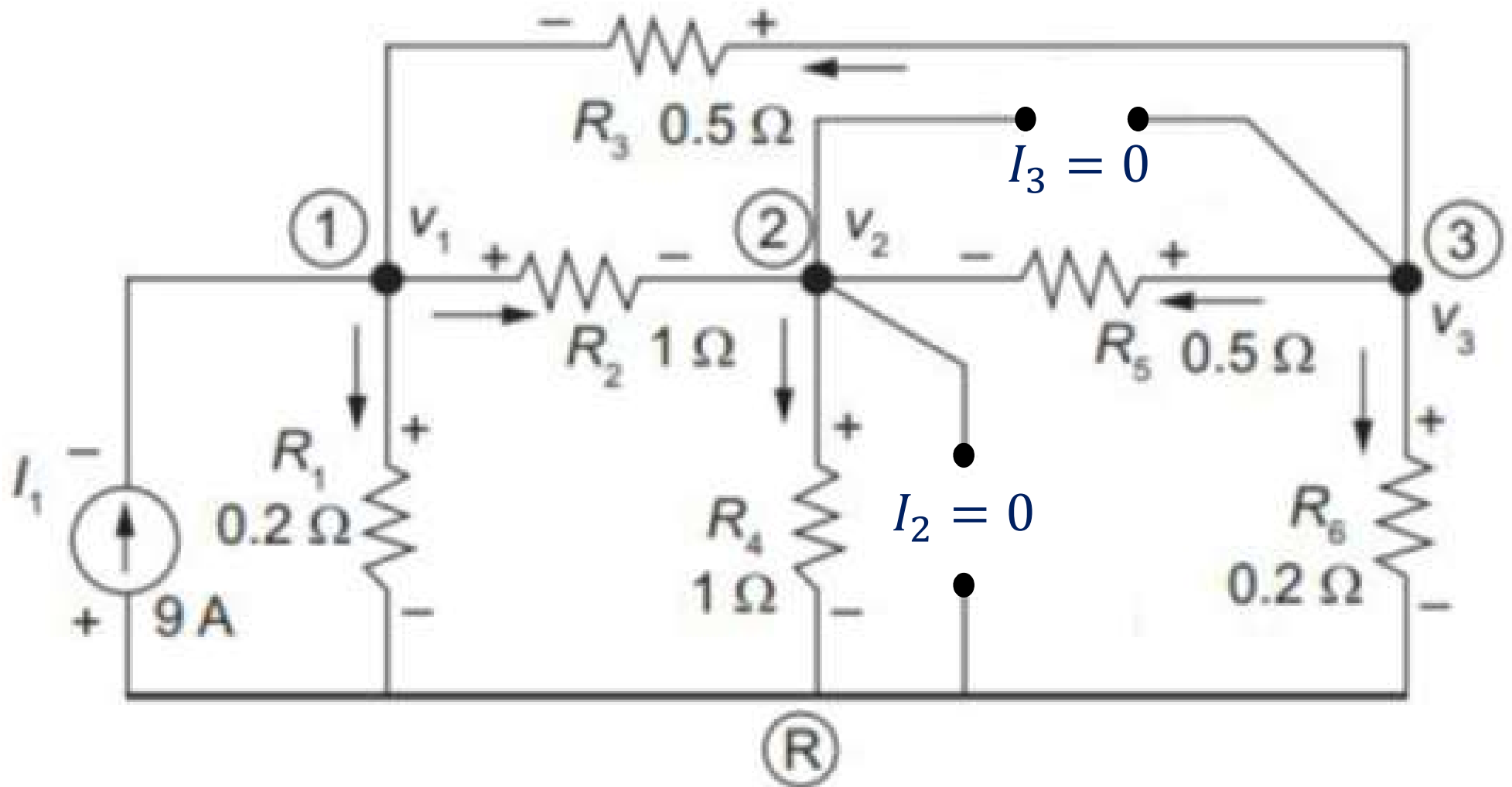


$$\begin{bmatrix} 8 & -1 & -2 \\ -1 & 4 & -2 \\ -2 & -2 & 9 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 9 \\ -17 \\ 21 \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

# Superposition Theorem

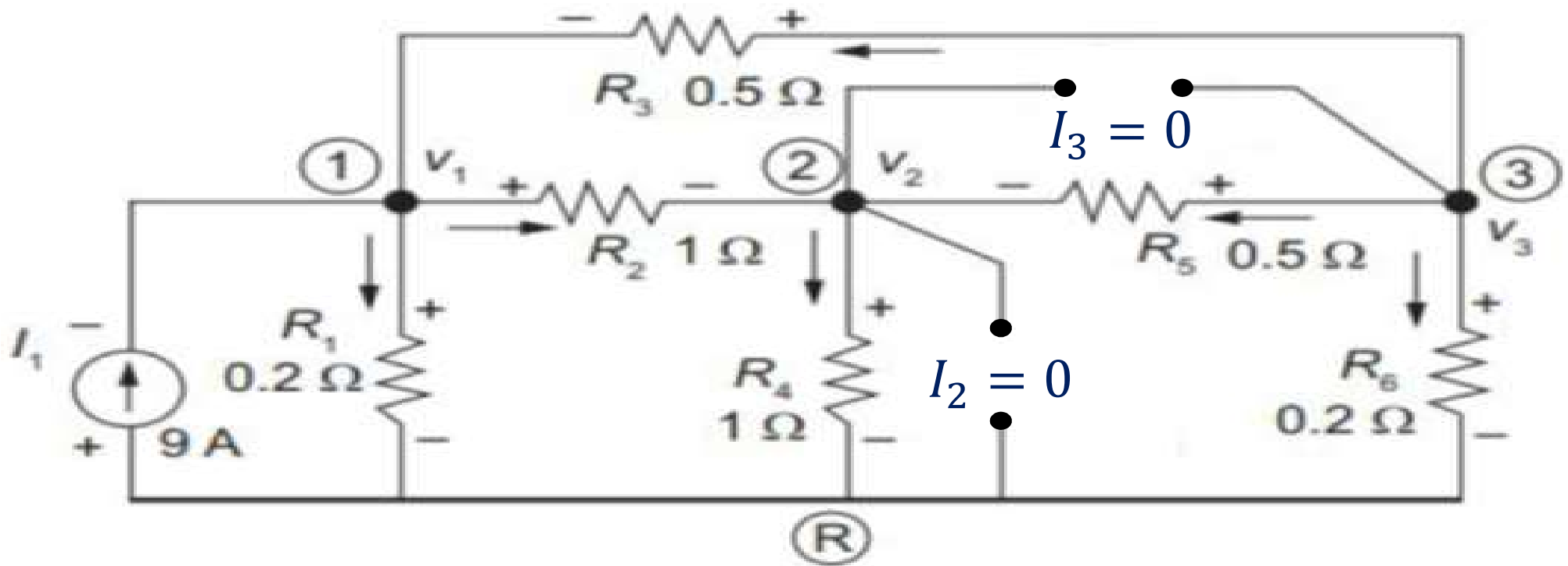
The response of any *Circuit Element* in a multi-source Resistive Circuit containing  $m$  *Independent Sources* can be obtained by *Adding the Responses* of the Same *Circuit Element* in  $m$  *Single Source Circuits*. Here, the  $i^{th}$  *Single Source Circuit* is formed by keeping only the  $i^{th}$  *Independent Source Active* while the *Remaining Sources are Deactivated*.

## Superposition Theorem: Only Source $I_1$ Active



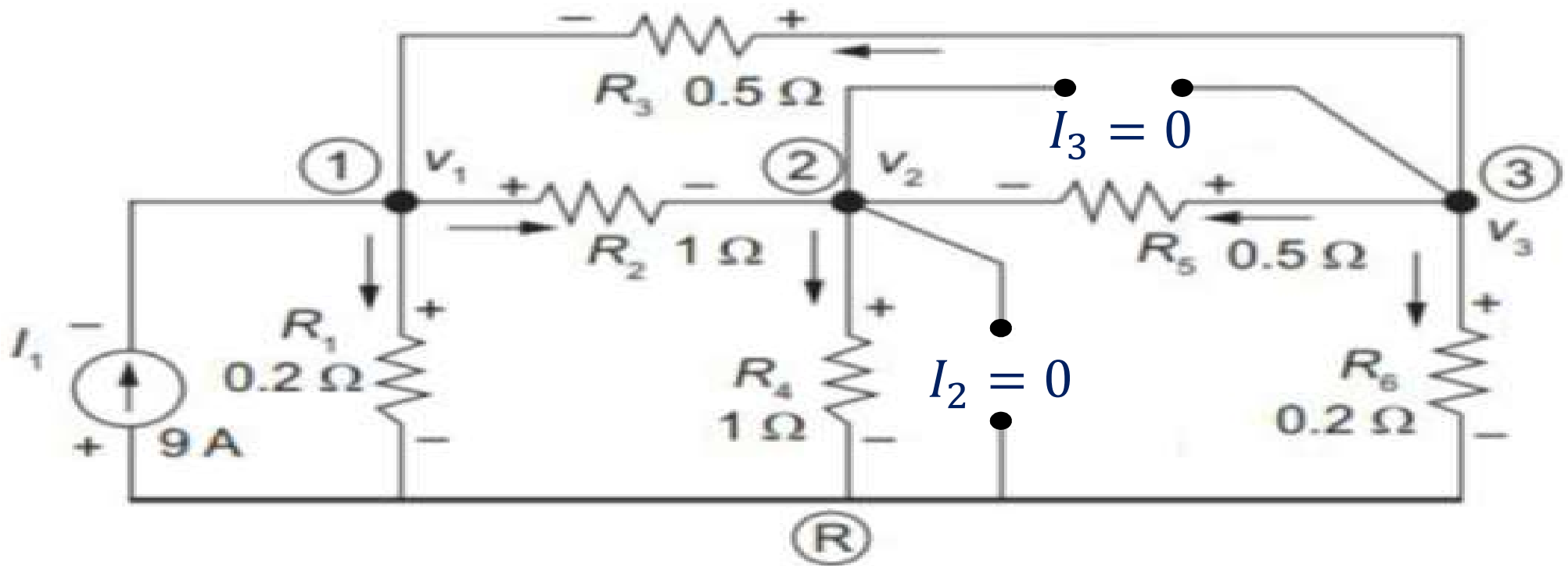


## Superposition Theorem: Only Source $I_1$ Active



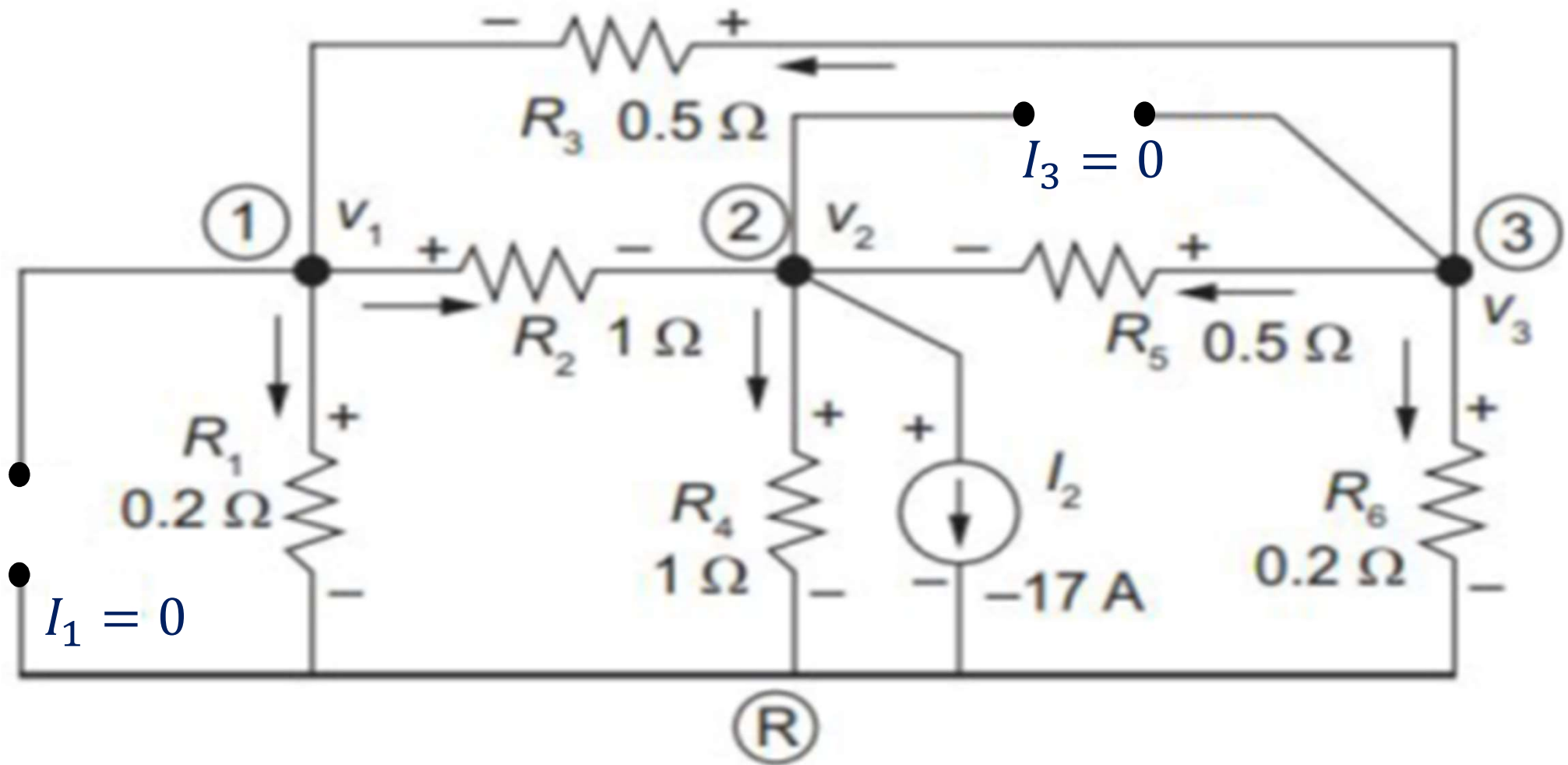
$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1(I_1) \\ V_2(I_1) \\ V_3(I_1) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ 0 \\ 0 \end{bmatrix}$$

## Superposition Theorem: Only Source $I_1$ Active

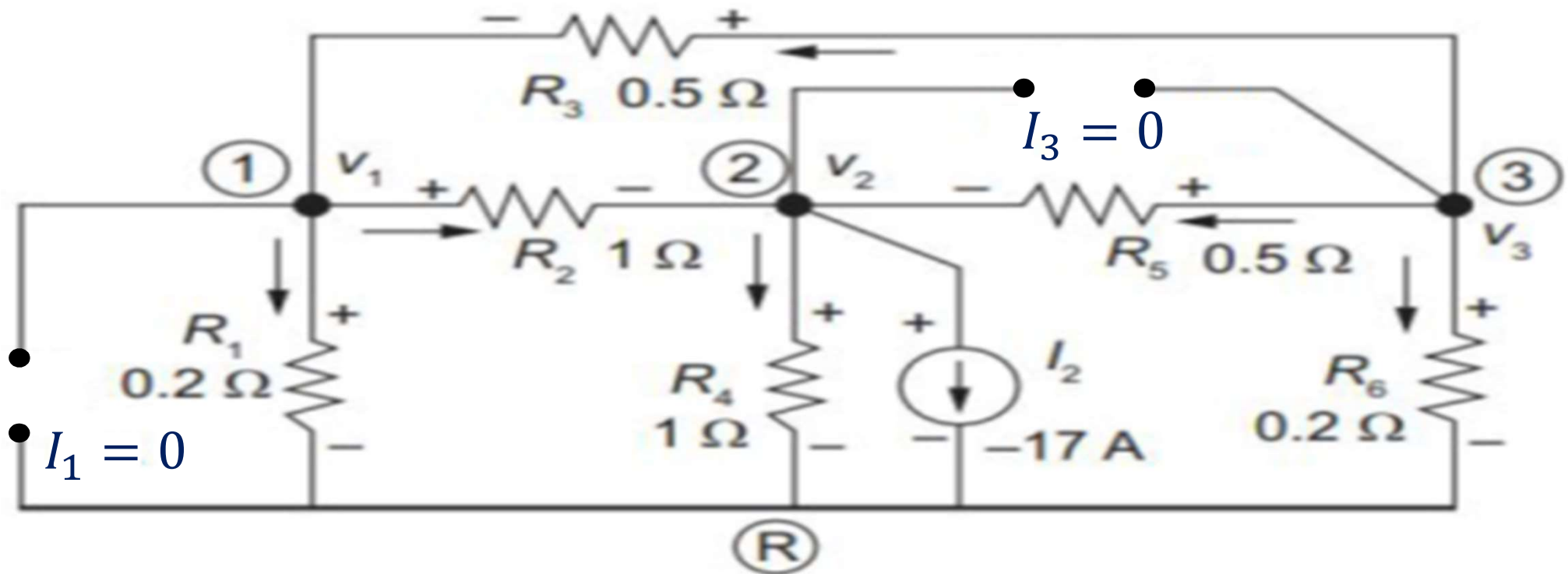


$$\begin{bmatrix} 8 & -1 & -2 \\ -1 & 4 & -2 \\ -2 & -2 & 9 \end{bmatrix} \begin{bmatrix} V_1(I_1) \\ V_2(I_1) \\ V_3(I_1) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} V_1(I_1) \\ V_2(I_1) \\ V_3(I_1) \end{bmatrix} = \begin{bmatrix} 1.291 \\ 0.525 \\ 0.404 \end{bmatrix}$$

## Superposition Theorem: Only Source $I_2$ Active

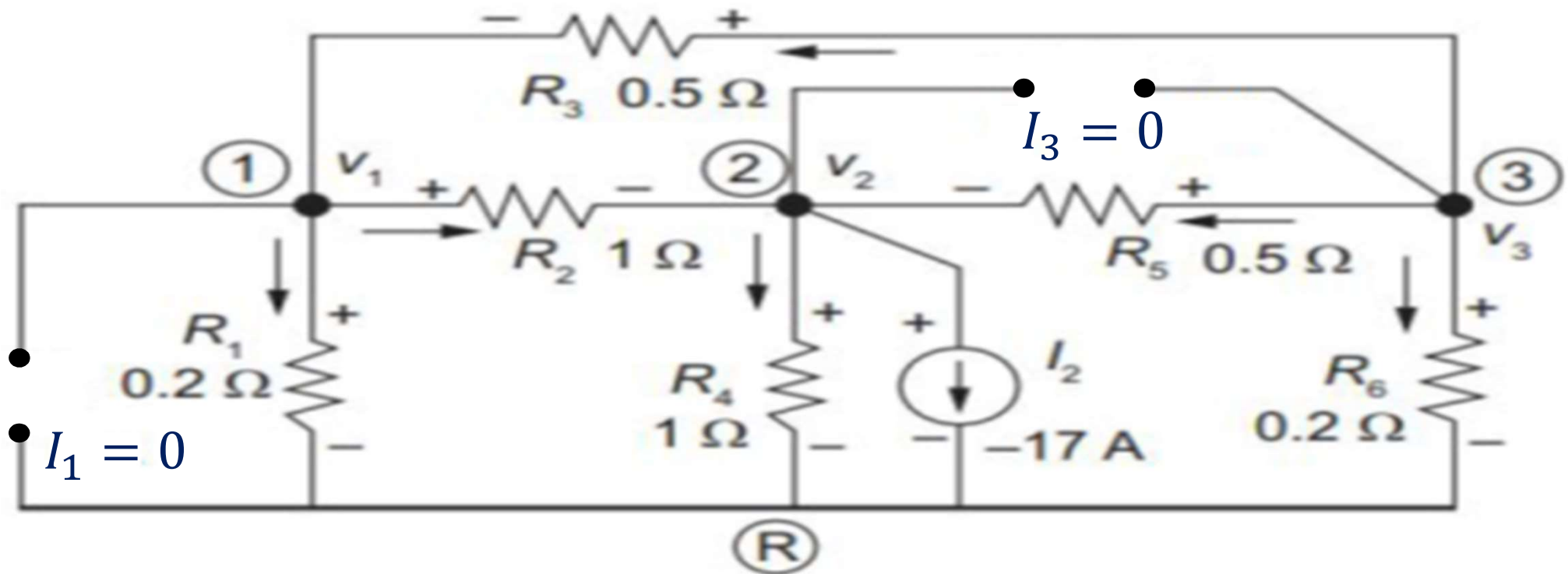


## Superposition Theorem: Only Source $I_2$ Active



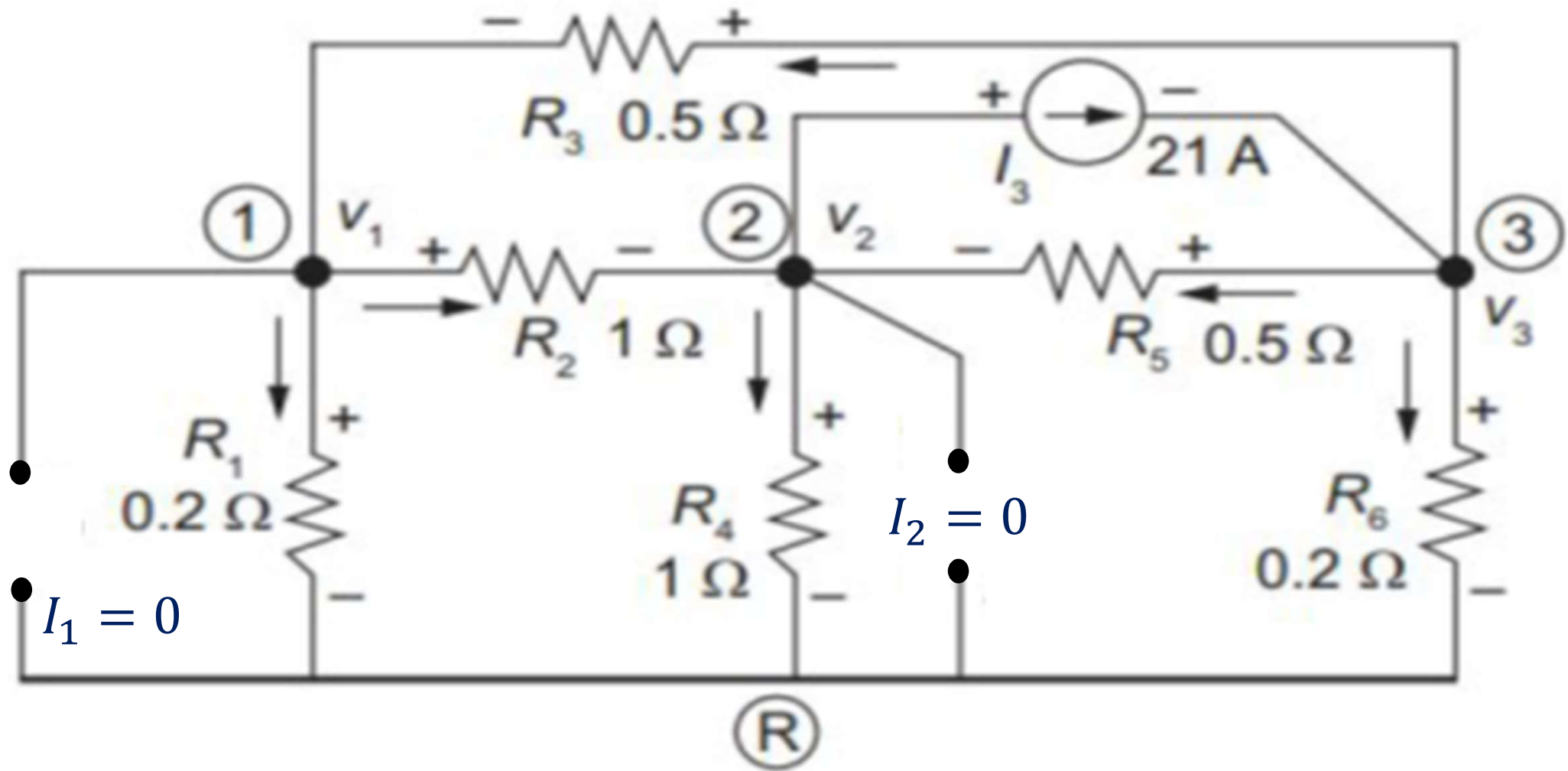
$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1(I_2) \\ V_2(I_2) \\ V_3(I_2) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ I_2 \\ 0 \end{bmatrix}$$

## Superposition Theorem: Only Source $I_2$ Active

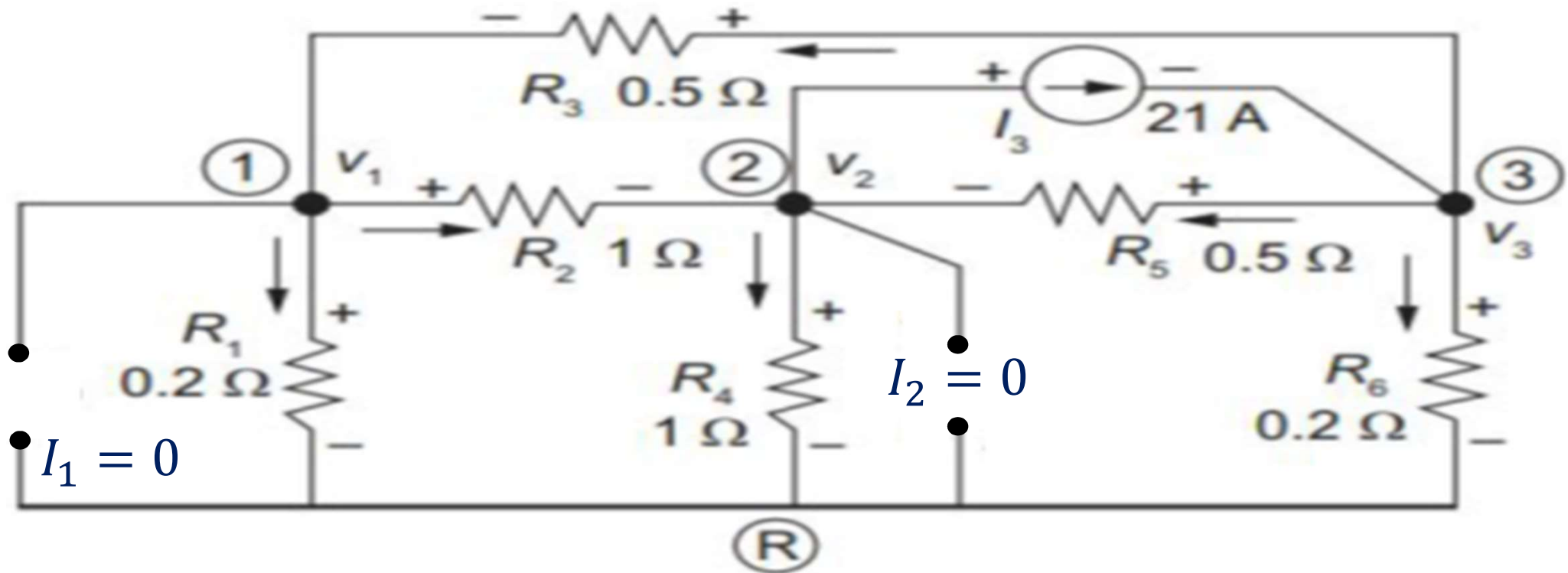


$$\begin{bmatrix} 8 & -1 & -2 \\ -1 & 4 & -2 \\ -2 & -2 & 9 \end{bmatrix} \begin{bmatrix} V_1(I_2) \\ V_2(I_2) \\ V_3(I_2) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -17 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} V_1(I_2) \\ V_2(I_2) \\ V_3(I_2) \end{bmatrix} = \begin{bmatrix} 0.991 \\ 5.184 \\ 1.372 \end{bmatrix}$$

Superposition Theorem: Only Source  $I_3$  Active

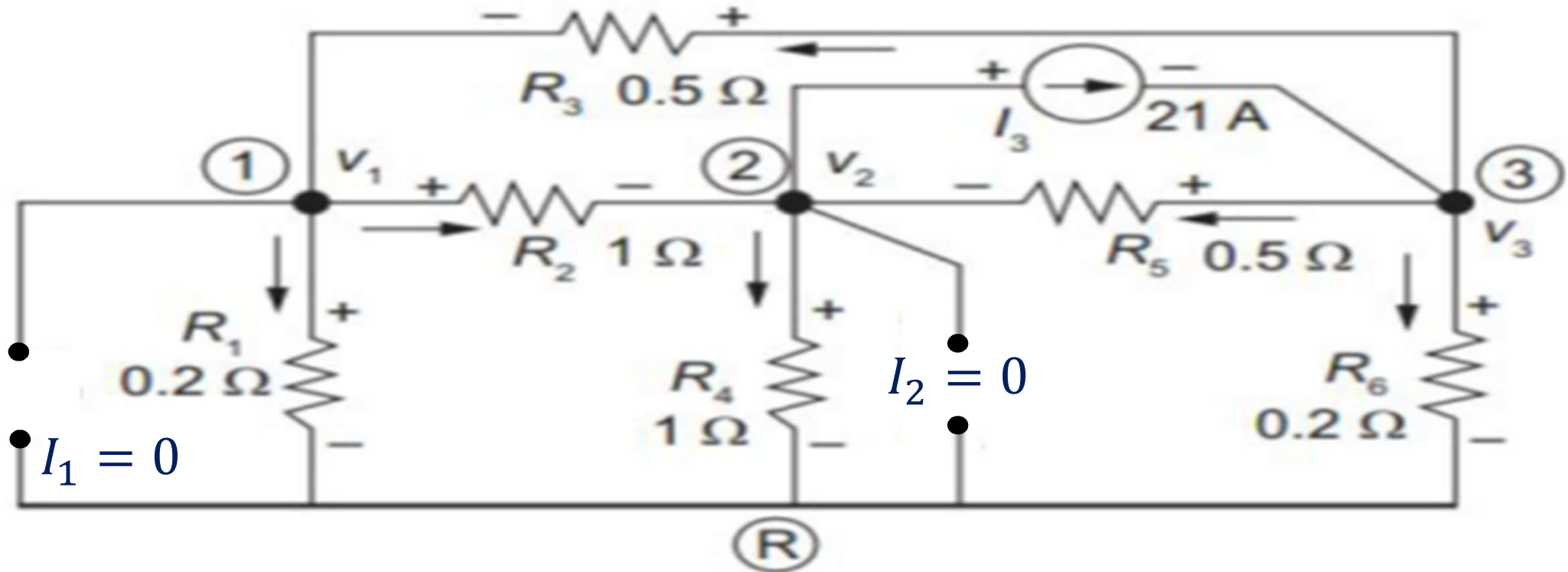


## Superposition Theorem: Only Source $I_3$ Active



$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_2 & -G_3 \\ -G_2 & G_2 + G_4 + G_5 & -G_5 \\ -G_3 & -G_5 & G_3 + G_5 + G_6 \end{bmatrix} \begin{bmatrix} V_1(I_3) \\ V_2(I_3) \\ V_3(I_3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ I_3 \end{bmatrix}$$

## Superposition Theorem: Only Source $I_3$ Active

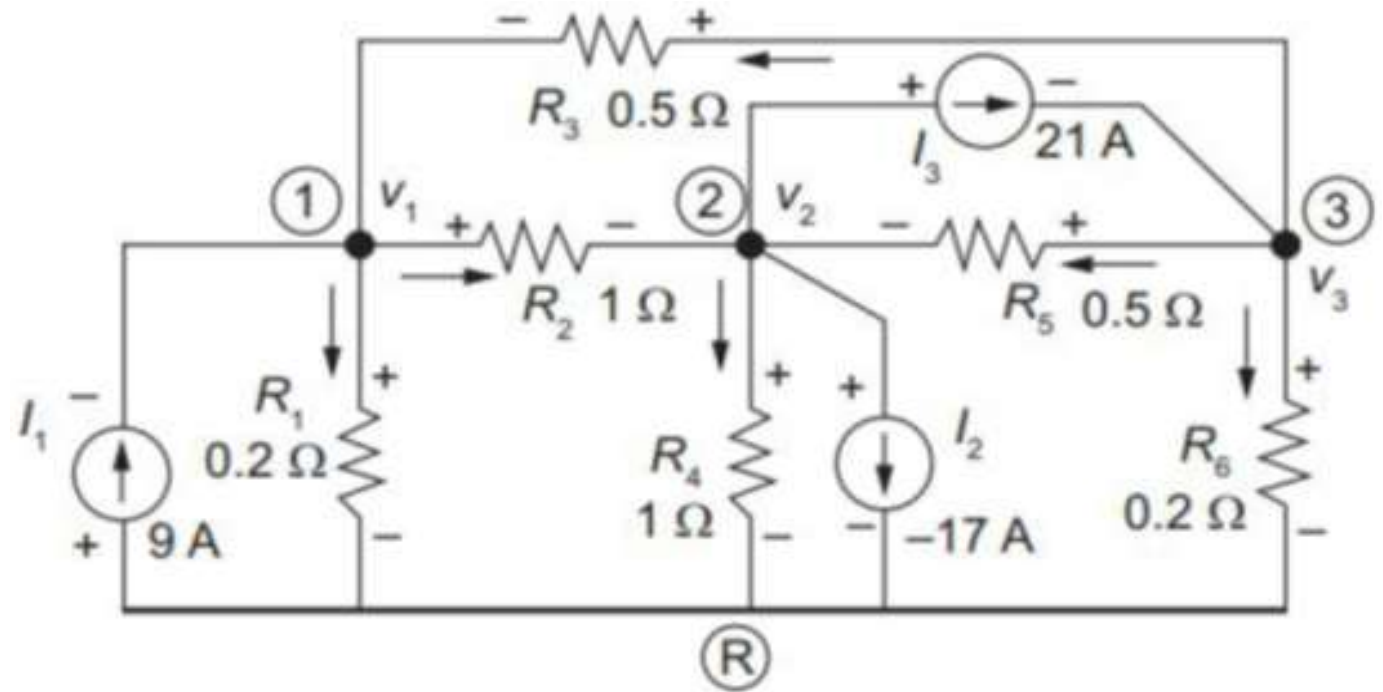


$$\begin{bmatrix} 8 & -1 & -2 \\ -1 & 4 & -2 \\ -2 & -2 & 9 \end{bmatrix} \begin{bmatrix} V_1(I_3) \\ V_2(I_3) \\ V_3(I_3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 21 \end{bmatrix} \Rightarrow \begin{bmatrix} V_1(I_3) \\ V_2(I_3) \\ V_3(I_3) \end{bmatrix} = \begin{bmatrix} -0.283 \\ -4.709 \\ 1.224 \end{bmatrix}$$



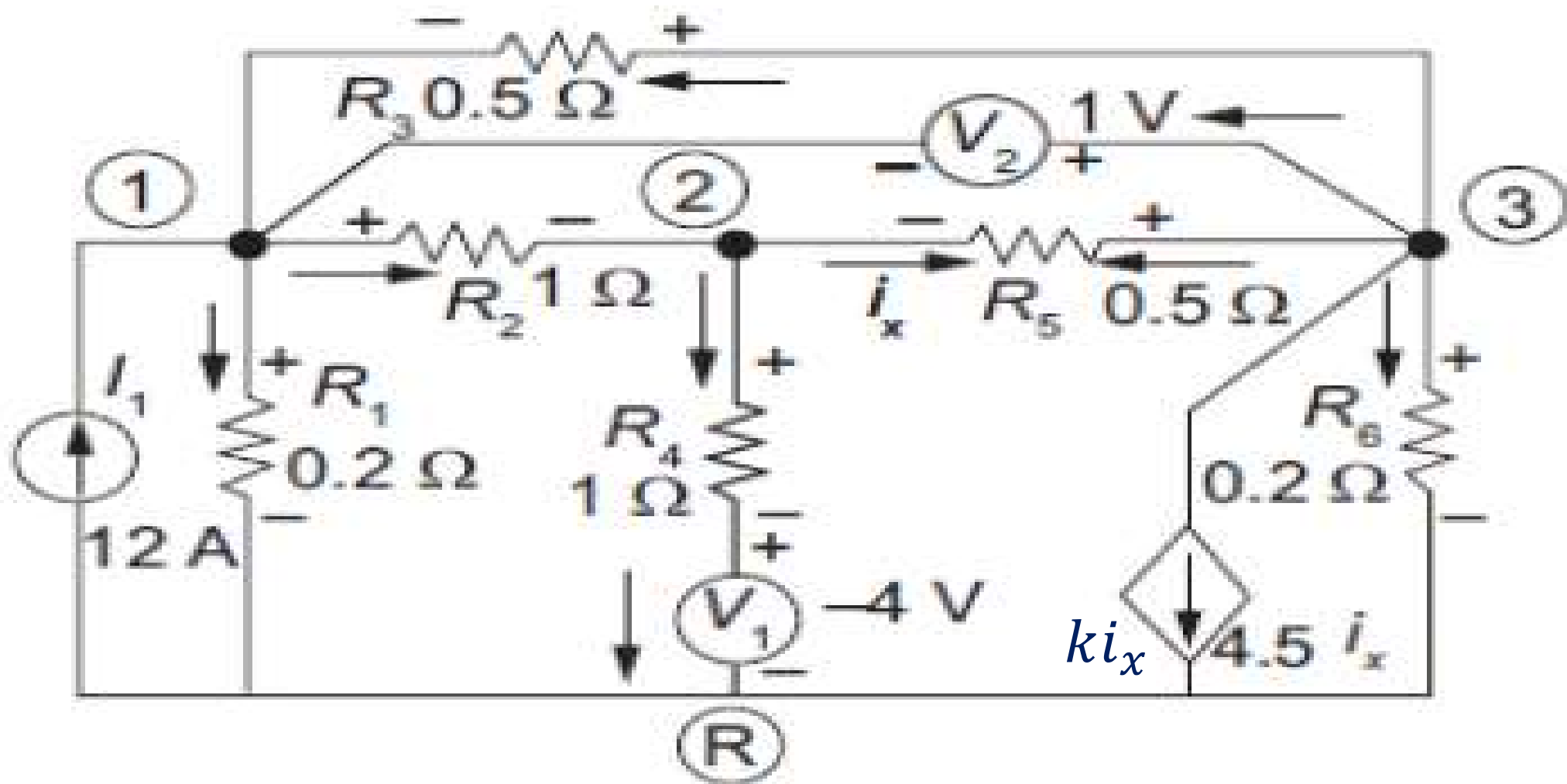
# Application of Superposition Theorem

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

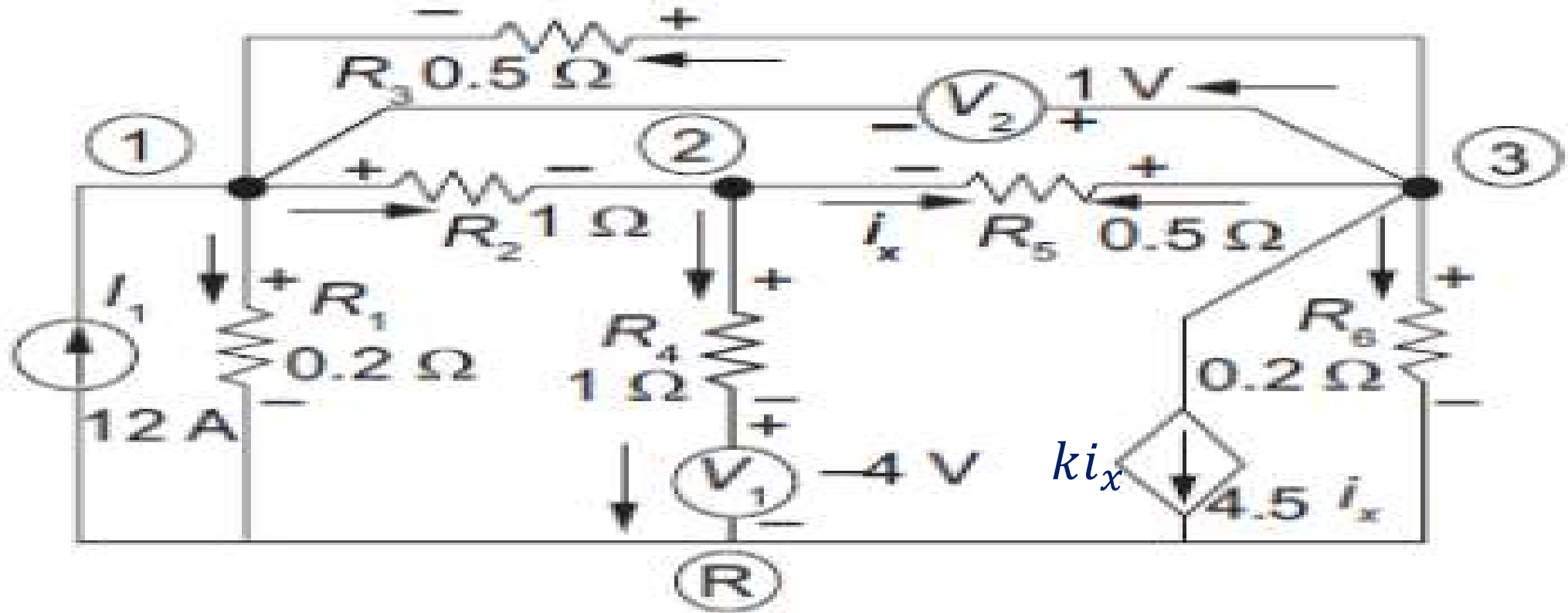


$$\begin{bmatrix} V_1(I_1) \\ V_2(I_1) \\ V_3(I_1) \end{bmatrix} + \begin{bmatrix} V_1(I_2) \\ V_2(I_2) \\ V_3(I_2) \end{bmatrix} + \begin{bmatrix} V_1(I_3) \\ V_2(I_3) \\ V_3(I_3) \end{bmatrix} = \begin{bmatrix} 1.291 \\ 0.525 \\ 0.404 \end{bmatrix} + \begin{bmatrix} 0.991 \\ 5.184 \\ 1.372 \end{bmatrix} + \begin{bmatrix} -0.283 \\ -4.709 \\ 1.224 \end{bmatrix} = \begin{bmatrix} 1.999 \\ 1.000 \\ 3.000 \end{bmatrix}$$

## Solution by Nodal Analysis

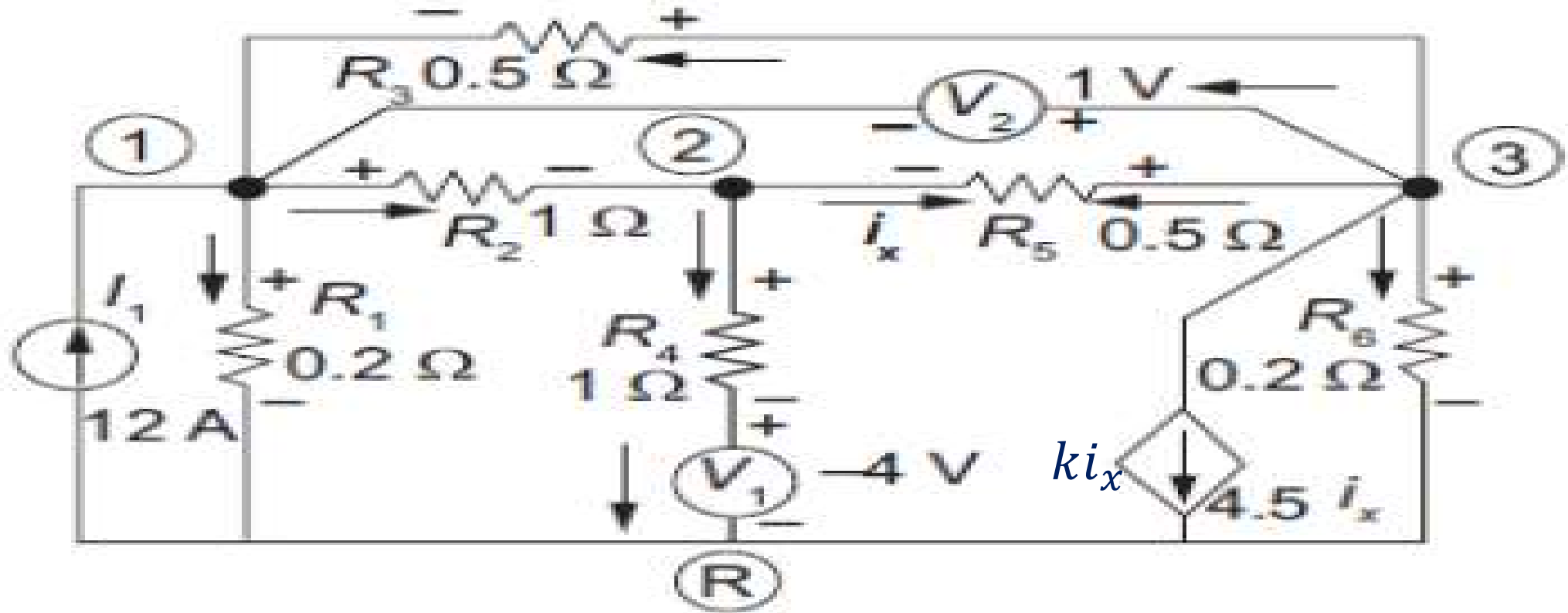


# Solution by Nodal Analysis

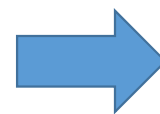


$$\begin{bmatrix} G_1 + G_2 + G_6 - (k-1)G_5 & (k-1)G_5 - G_2 \\ -(G_2 + G_5) & G_2 + G_4 + G_5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & (k-1)G_5 - G_6 \\ 0 & G_4 & G_5 \end{bmatrix} \begin{bmatrix} I_1 \\ V_1 \\ V_2 \end{bmatrix}$$

# Solution by Nodal Analysis

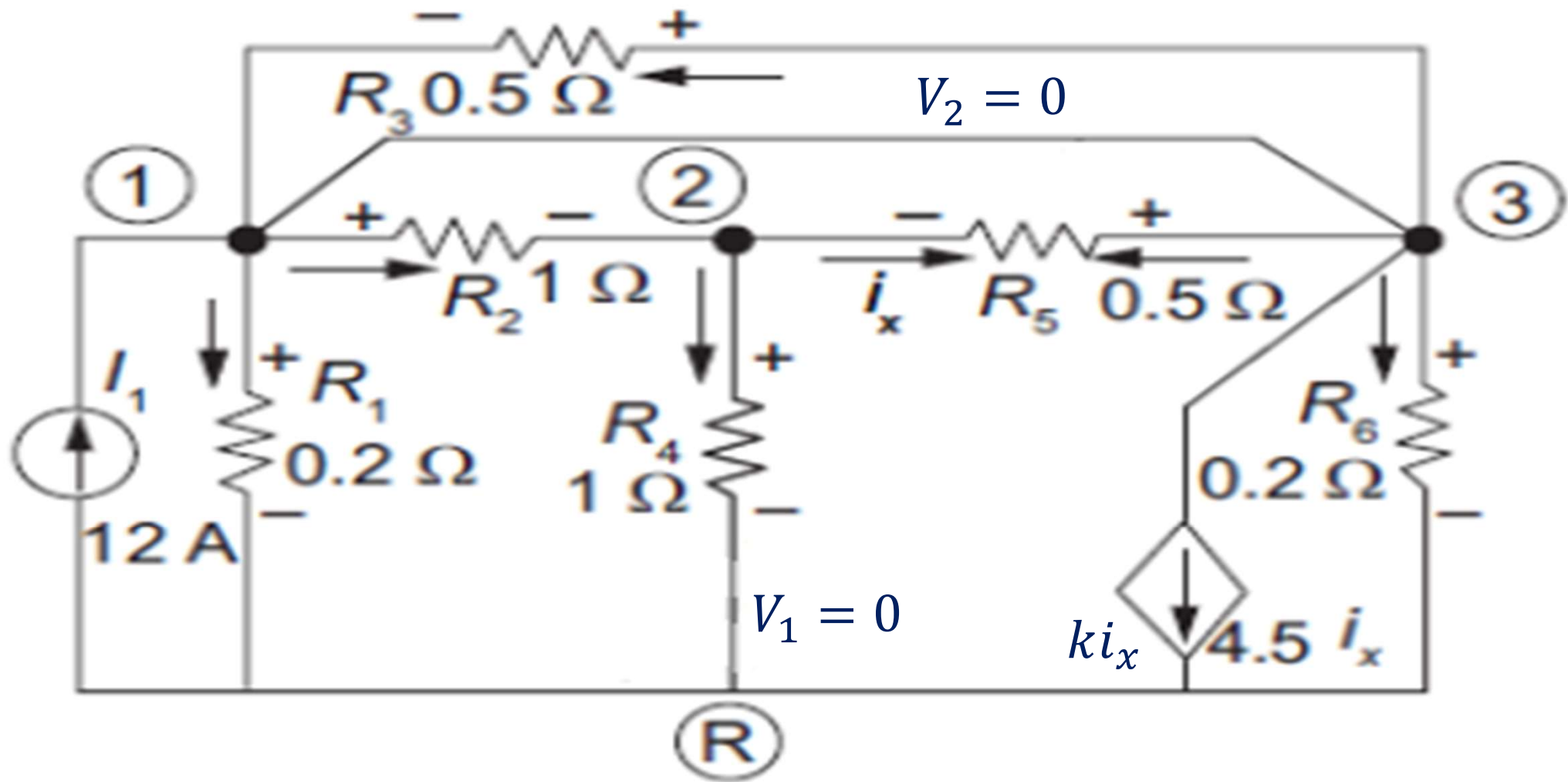


$$\begin{bmatrix} 4 & 6 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 12 \\ -4 \\ 1 \end{bmatrix}$$

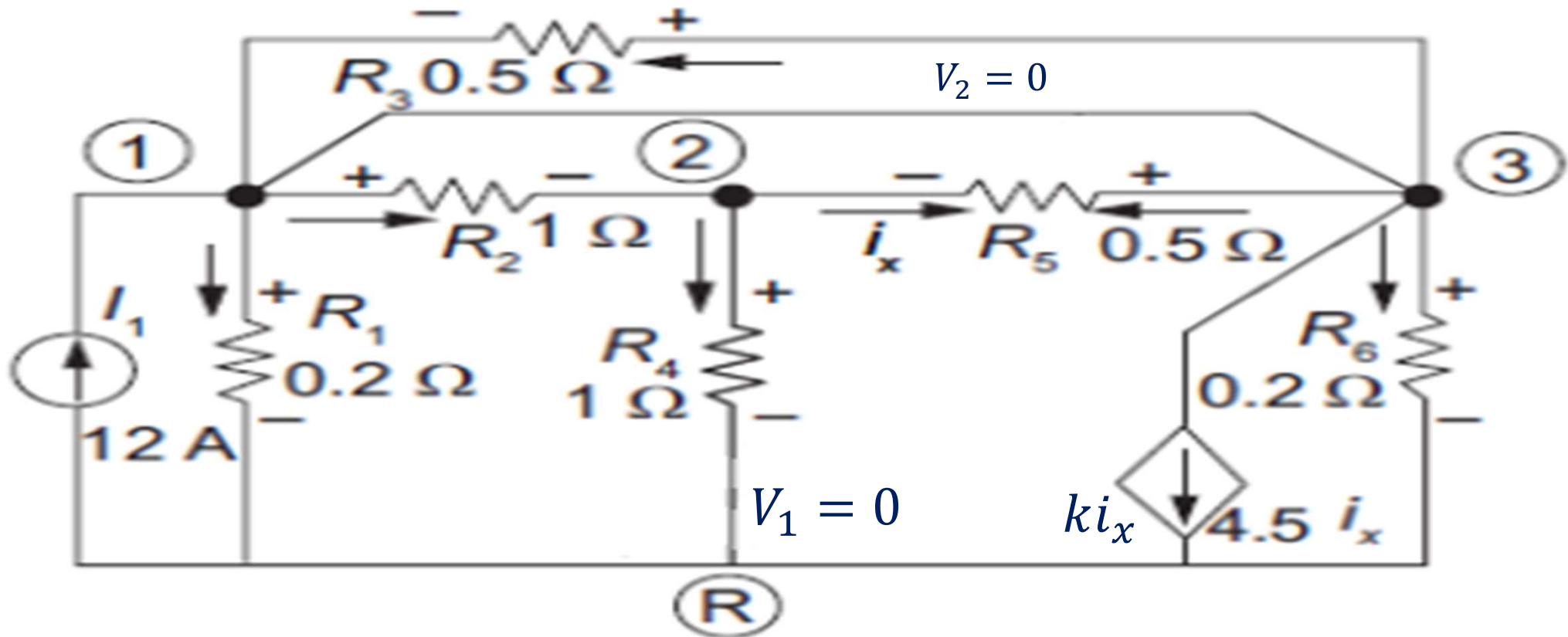


$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2.000 \\ 1.000 \end{bmatrix}$$

Superposition Theorem: Only Source  $I_1$  Active

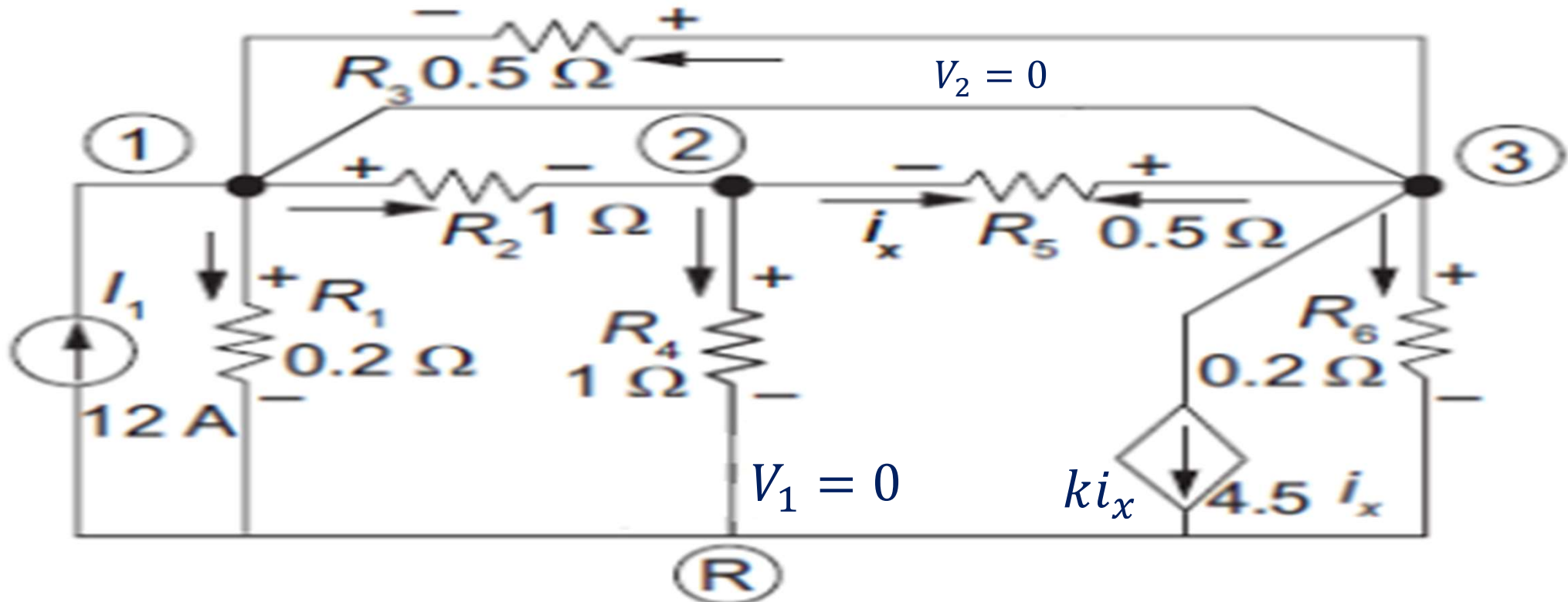


## Superposition Theorem: Only Source $I_1$ Active



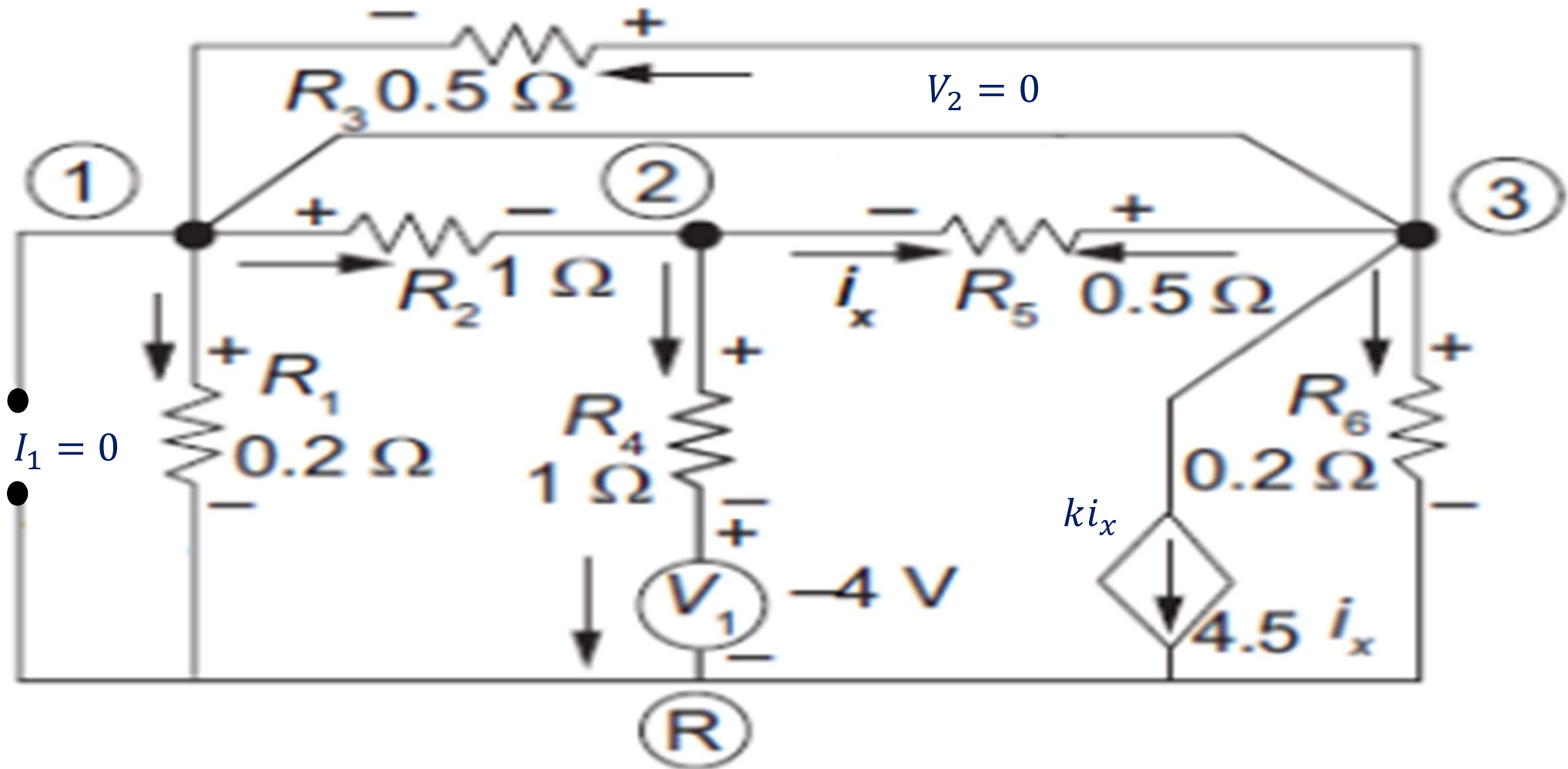
$$\begin{bmatrix} G_1 + G_2 + G_6 - (k-1)G_5 & (k-1)G_5 - G_2 \\ -(G_2 + G_5) & G_2 + G_4 + G_5 \end{bmatrix} \begin{bmatrix} v_1(I_1) \\ v_2(I_1) \end{bmatrix} = \begin{bmatrix} 1 & 0 & (k-1)G_5 - G_6 \\ 0 & G_4 & G_5 \end{bmatrix} \begin{bmatrix} I_1 \\ 0 \\ 0 \end{bmatrix}$$

## Superposition Theorem: Only Source $I_1$ Active



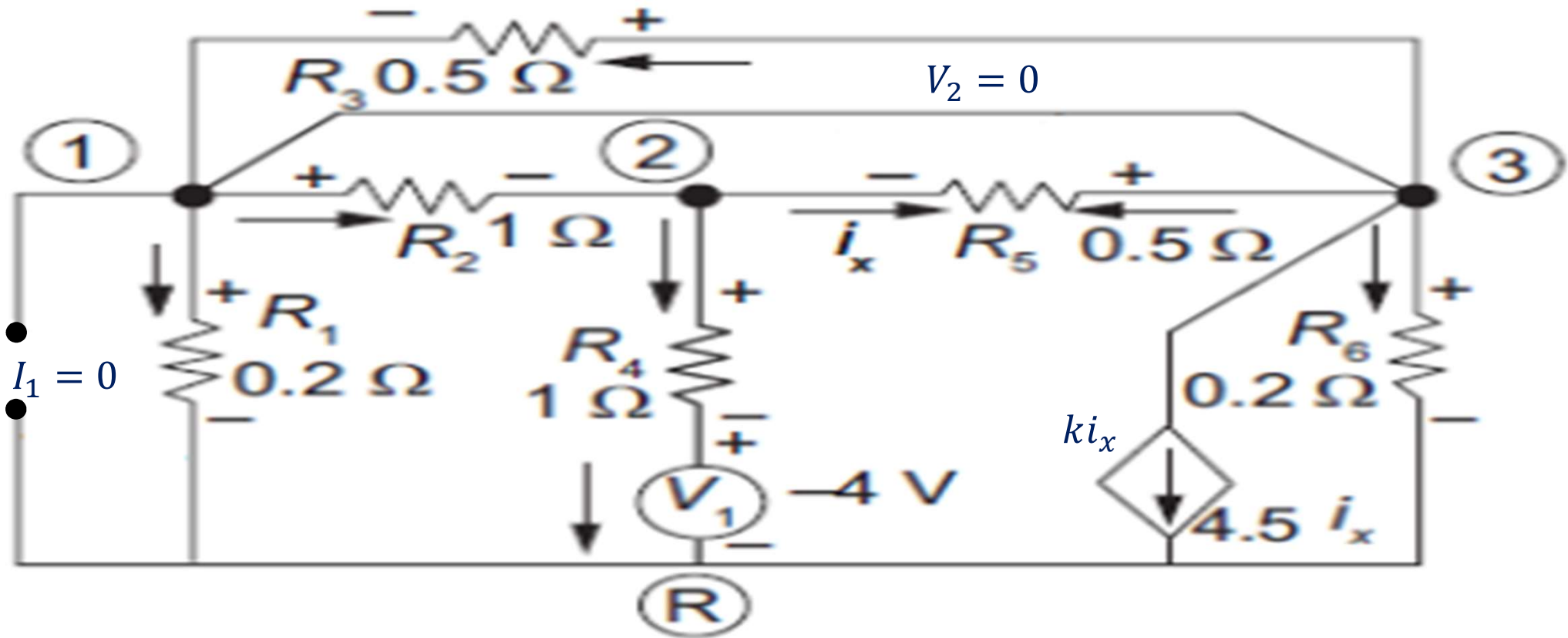
$$\begin{bmatrix} 4 & 6 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} v_1(I_1) \\ v_2(I_1) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} v_1(I_1) \\ v_2(I_1) \end{bmatrix} = \begin{bmatrix} 1.412 \\ 1.059 \end{bmatrix}$$

## Superposition Theorem: Only Source $V_1$ Active



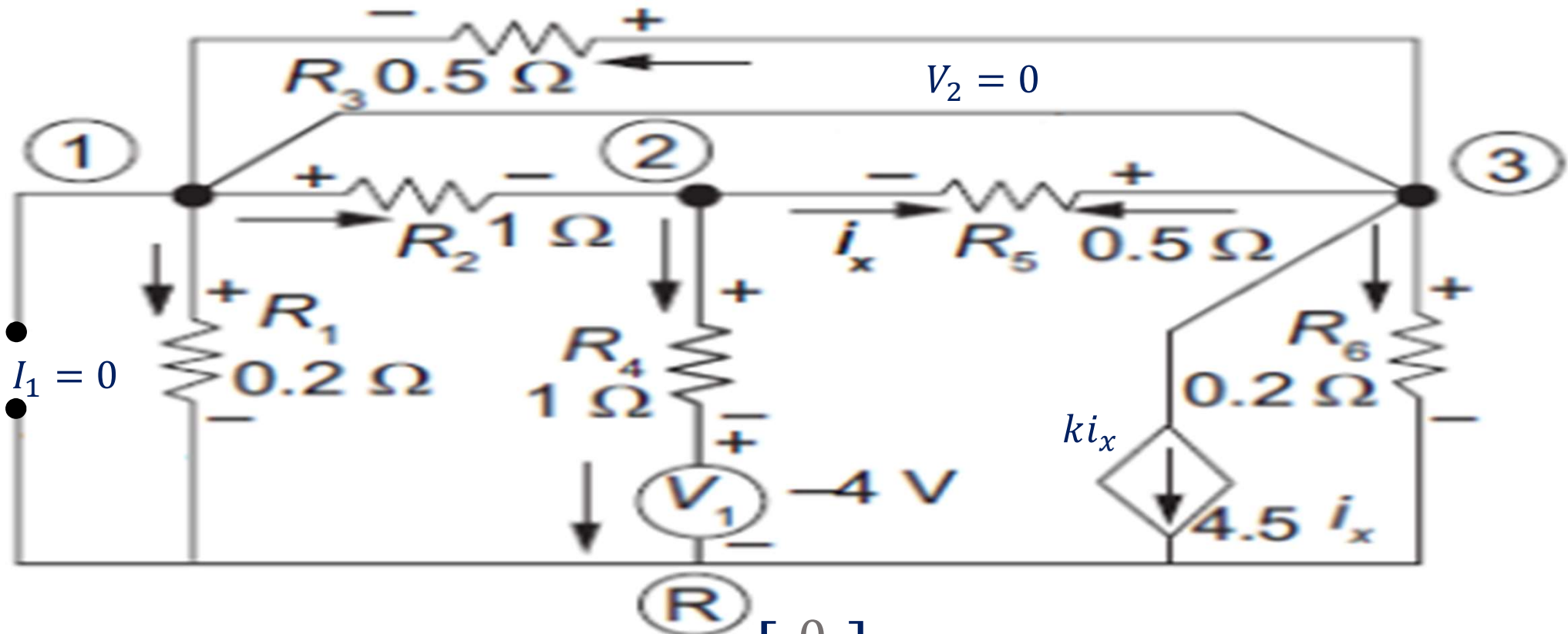


## Superposition Theorem: Only Source $V_1$ Active



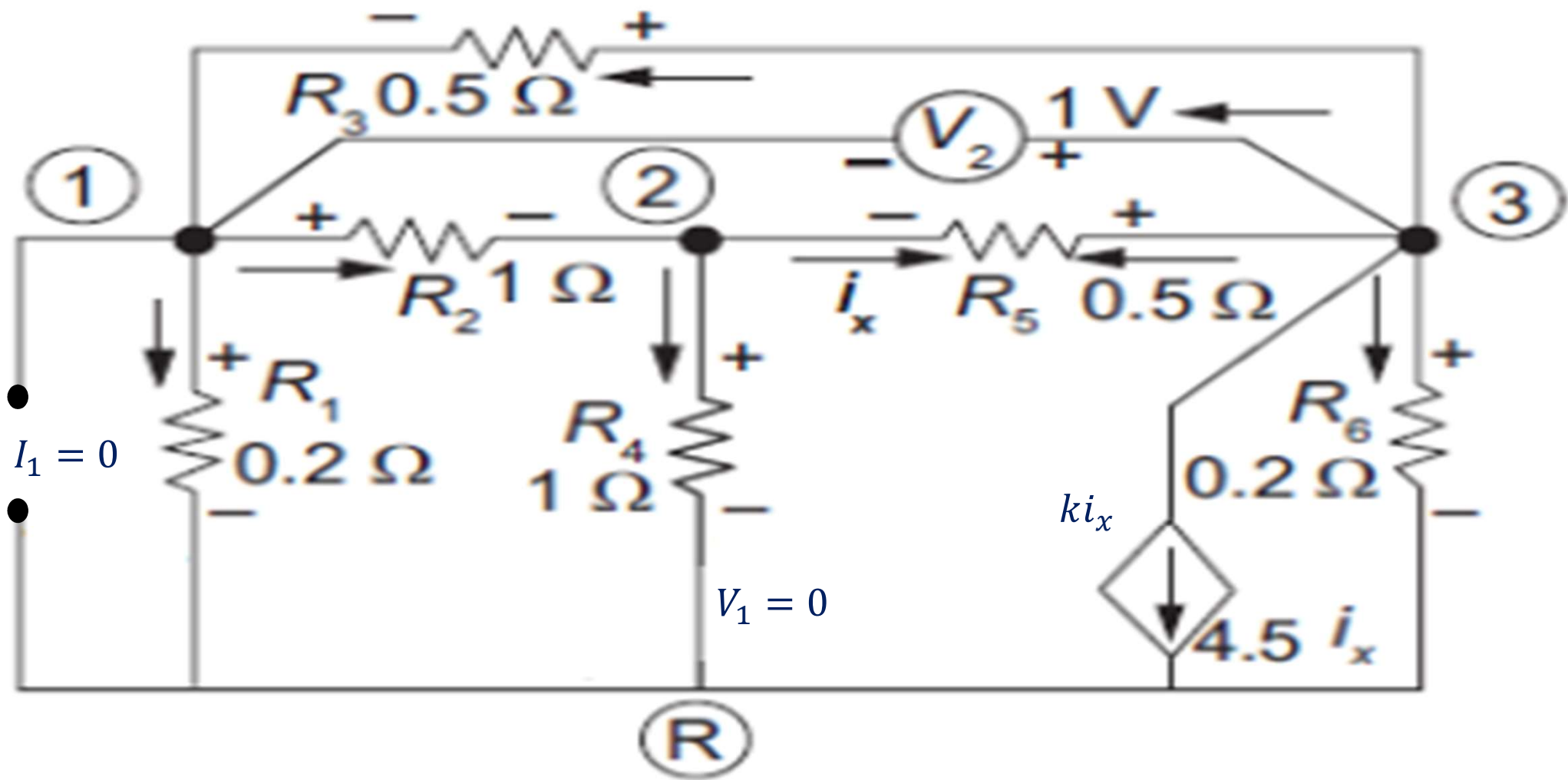
$$\begin{bmatrix} G_1 + G_2 + G_6 - (k - 1)G_5 & (k - 1)G_5 - G_2 \\ -(G_2 + G_5) & G_2 + G_4 + G_5 \end{bmatrix} \begin{bmatrix} v_1(V_1) \\ v_2(V_1) \end{bmatrix} = \begin{bmatrix} 1 & 0 & (k - 1)G_5 - G_6 \\ 0 & G_4 & G_5 \end{bmatrix} \begin{bmatrix} 0 \\ V_1 \\ 0 \end{bmatrix}$$

## Superposition Theorem: Only Source $V_1$ Active

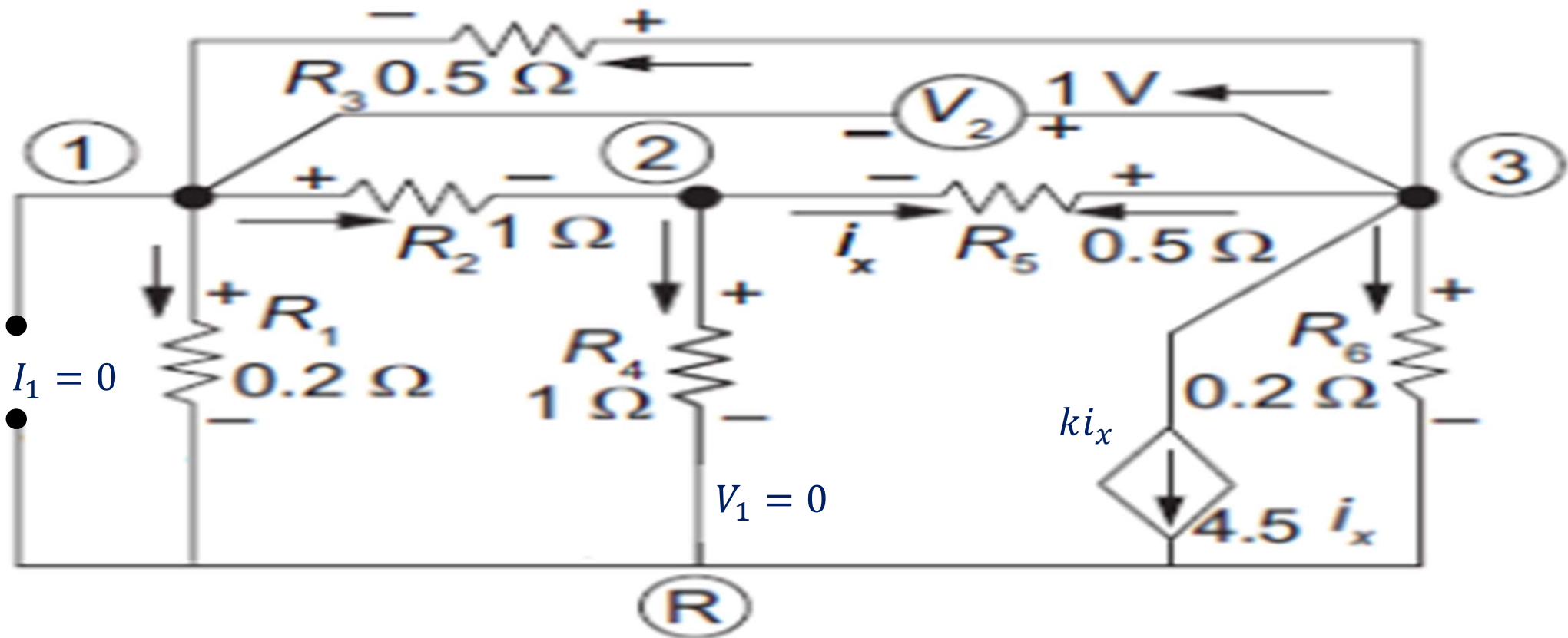


$$\begin{bmatrix} 4 & 6 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} v_1(V_1) \\ v_2(V_1) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ -4 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} v_1(V_1) \\ v_2(V_1) \end{bmatrix} = \begin{bmatrix} 0.706 \\ -0.471 \end{bmatrix}$$

Superposition Theorem: Only Source  $V_2$  Active

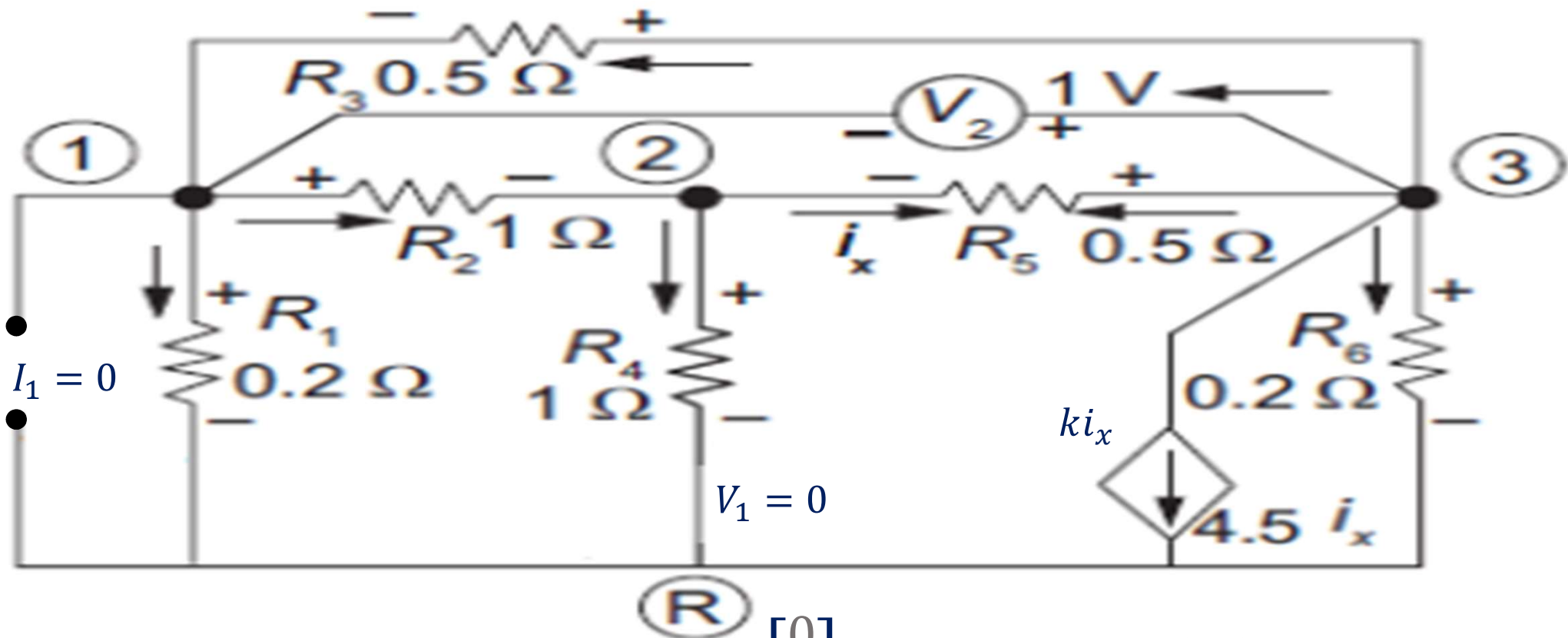


## Superposition Theorem: Only Source $V_2$ Active



$$\begin{bmatrix} G_1 + G_2 + G_6 - (k-1)G_5 & (k-1)G_5 - G_2 \\ -(G_2 + G_5) & G_2 + G_4 + G_5 \end{bmatrix} \begin{bmatrix} v_1(V_2) \\ v_2(V_2) \end{bmatrix} = \begin{bmatrix} 1 & 0 & (k-1)G_5 - G_6 \\ 0 & G_4 & G_5 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ V_2 \end{bmatrix}$$

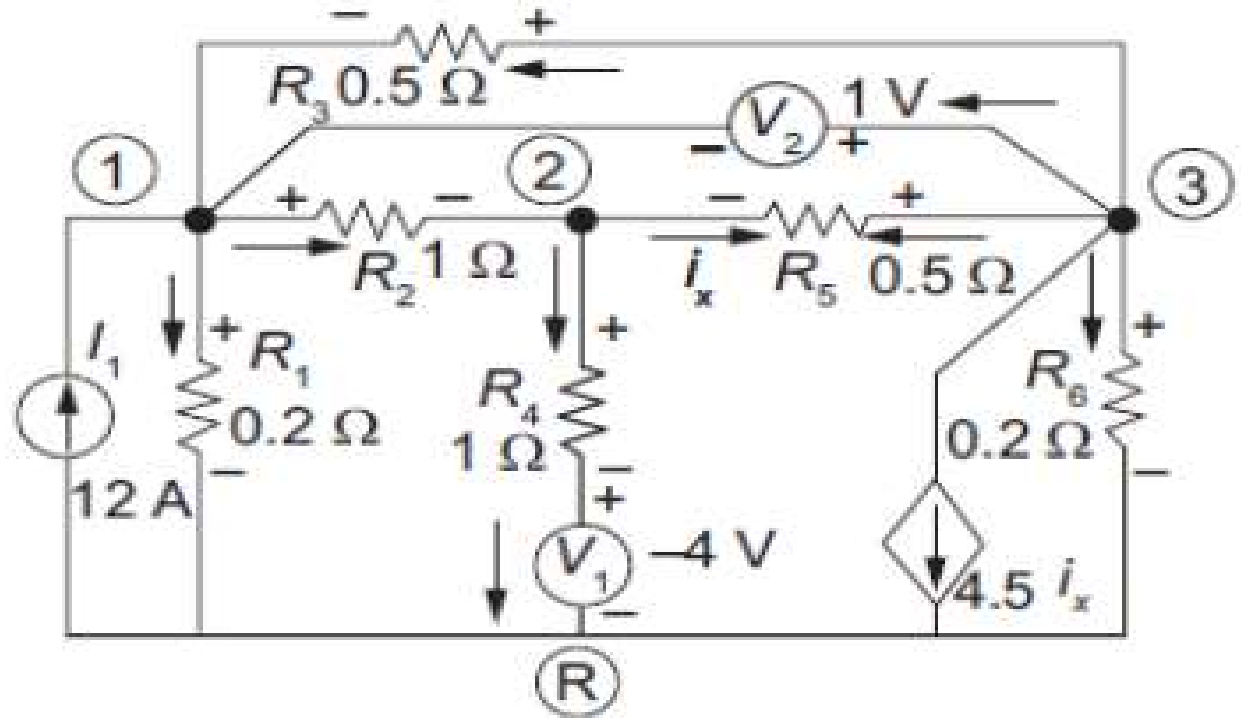
## Superposition Theorem: Only Source $V_2$ Active



$$\begin{bmatrix} 4 & 6 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} v_1(V_2) \\ v_2(V_2) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} v_1(V_2) \\ v_2(V_2) \end{bmatrix} = \begin{bmatrix} -0.118 \\ 0.412 \end{bmatrix}$$

# Application of Superposition Theorem

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2.000 \\ 1.000 \end{bmatrix}$$



$$\begin{bmatrix} v_1(I_1) \\ v_2(I_1) \end{bmatrix} + \begin{bmatrix} v_1(V_1) \\ v_2(V_1) \end{bmatrix} + \begin{bmatrix} v_1(V_2) \\ v_2(V_2) \end{bmatrix} = \begin{bmatrix} 1.412 \\ 1.059 \end{bmatrix} + \begin{bmatrix} 0.706 \\ -0.471 \end{bmatrix} + \begin{bmatrix} -0.118 \\ 0.412 \end{bmatrix} = \begin{bmatrix} 2.000 \\ 1.000 \end{bmatrix}$$

# Superposition Theorem

Consider an Electric Circuit with  $n$  nodes and  $m = m_{ics} + m_{ivs}$  independent sources  $u_1, \dots, u_j, \dots, u_m$ . We can write the following expression through nodal analysis to solve the node voltages  $v_1, \dots, v_k, \dots, v_q$  ( $q = n - 1 - m_{ivs}$ ).

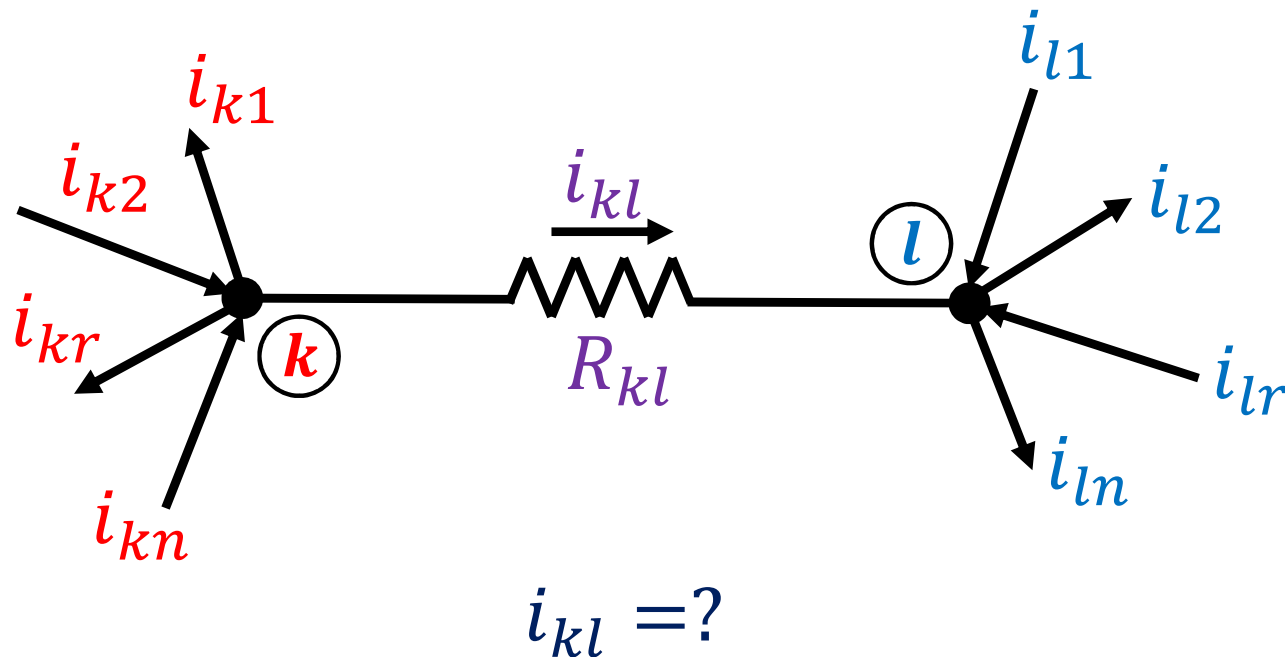
$$Y_{q \times q} V_{q \times 1} = C_{q \times m} U_{m \times 1}$$



$$V = \begin{bmatrix} v_1 \\ \vdots \\ v_k \\ \vdots \\ v_q \end{bmatrix} = Y^{-1} C U \longrightarrow U = \begin{bmatrix} u_1 \\ \vdots \\ u_j \\ \vdots \\ u_m \end{bmatrix}$$

# Superposition Theorem

Consider measuring the current  $i_{kl}$  through the resistive branch (with resistance  $R_{kl}$ ) connecting the nodes  $k$  and  $l$ .

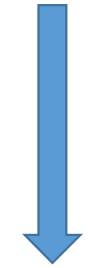




# Superposition Theorem

$$i_{kl} = \frac{v_k - v_l}{R_{kl}} = G_{kl} [0 \quad \dots \quad \boxed{1}_k \quad \dots \quad \boxed{-1}_l \quad \dots \quad 0] \begin{bmatrix} v_1 \\ \vdots \\ v_k \\ \vdots \\ v_l \\ \vdots \\ v_{n-1} \end{bmatrix}$$

$M_{kl}$



$$i_{kl} = M_{kl} V = M_{kl} Y^{-1} C U$$

# Superposition Theorem

$$i_{kl} = M_{kl}Y^{-1}CU = M_{kl}Y^{-1}C \begin{bmatrix} u_1 \\ \vdots \\ u_j \\ \vdots \\ u_m \end{bmatrix} = M_{kl}Y^{-1}C \left( \begin{bmatrix} u_1 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \cdots + \begin{bmatrix} 0 \\ \vdots \\ u_j \\ \vdots \\ 0 \end{bmatrix} + \cdots \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \vdots \\ u_m \end{bmatrix} \right)$$

$$i_{kl} = M_{kl}Y^{-1}C \mathbf{U}_1 + \cdots + M_{kl}Y^{-1}C \mathbf{U}_j + \cdots + M_{kl}Y^{-1}C \mathbf{U}_m$$

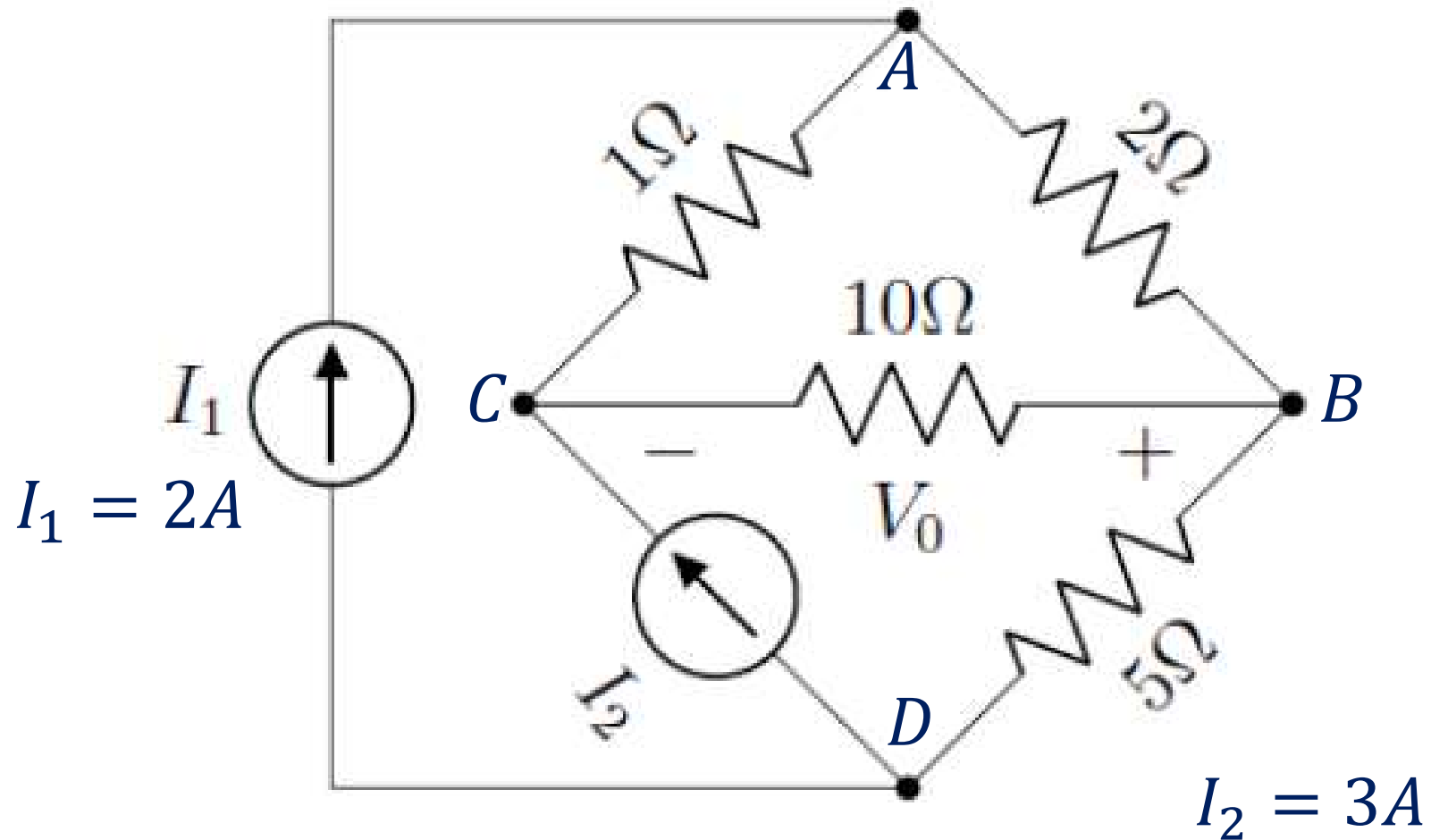
Only  $u_1$  Active

Only  $u_j$  Active

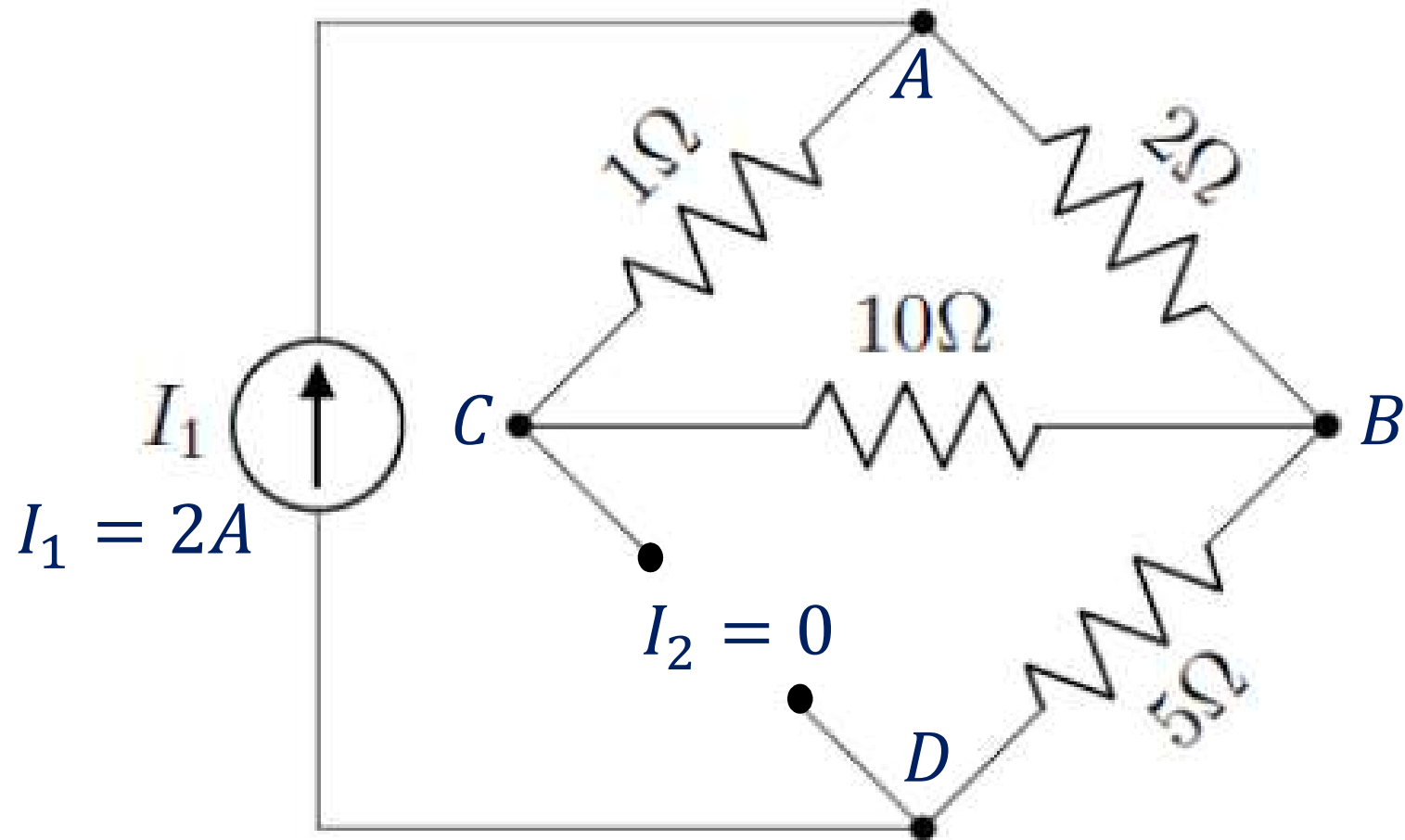
Only  $u_m$  Active

$$i_{kl} = i_{kl}(\mathbf{U}_1) + \cdots + i_{kl}(\mathbf{U}_j) + \cdots + i_{kl}(\mathbf{U}_m)$$

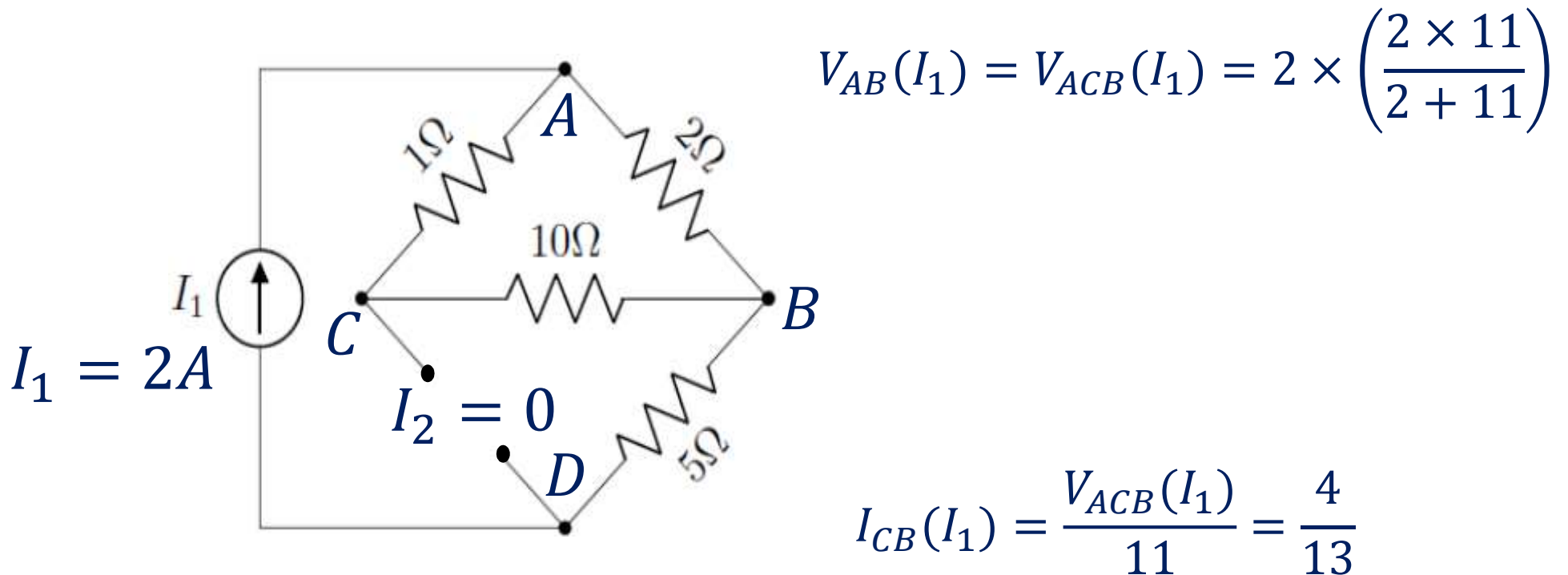
Solve for  $V_{BC}$  Using Superposition Theorem



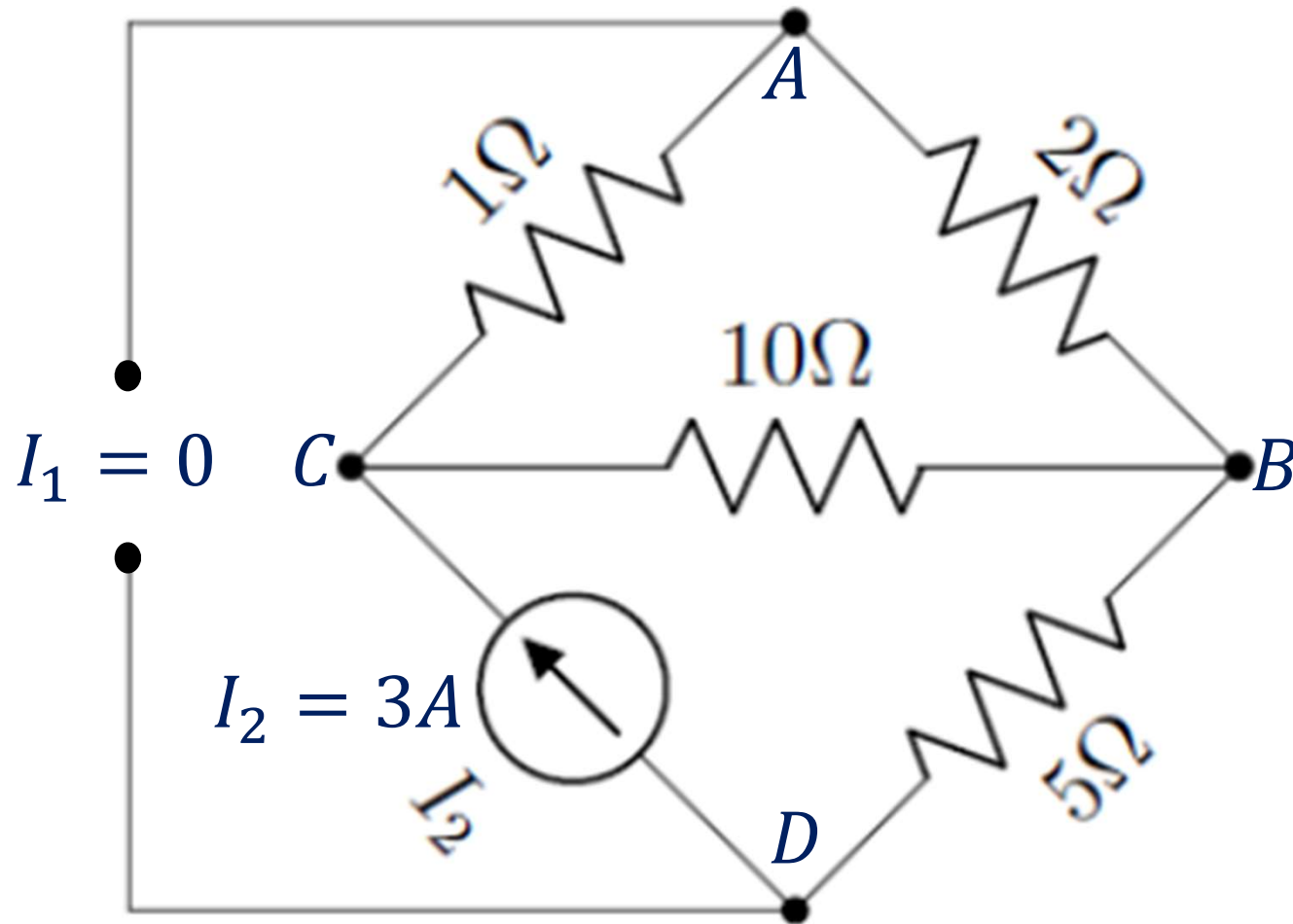
Superposition Theorem: Only  $I_1$  Active



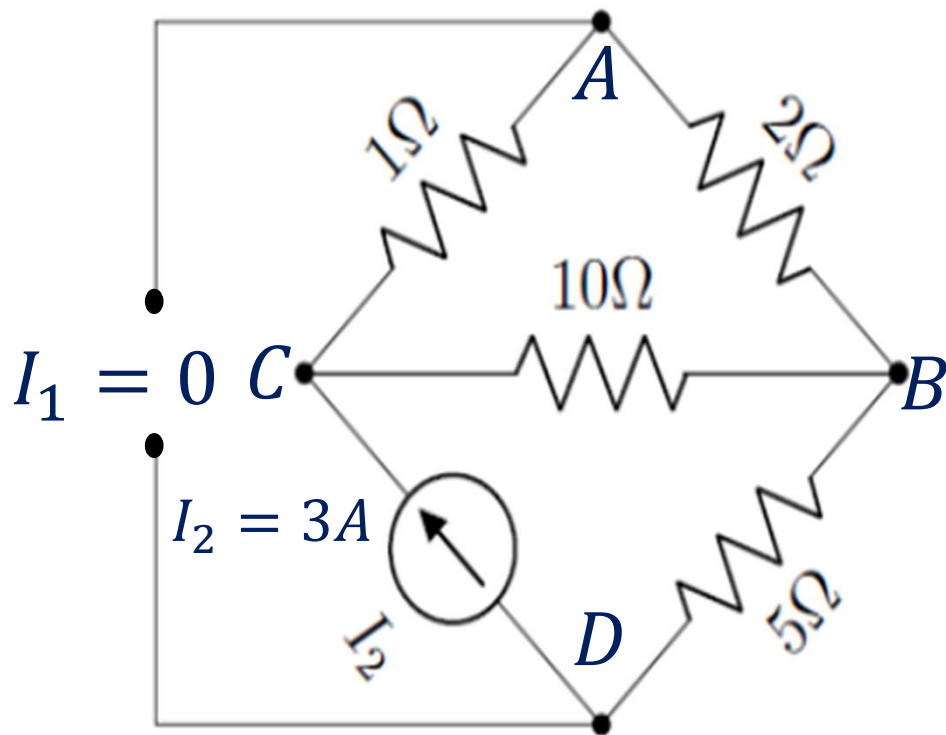
## Superposition Theorem: Only $I_1$ Active



## Superposition Theorem: Only $I_2$ Active



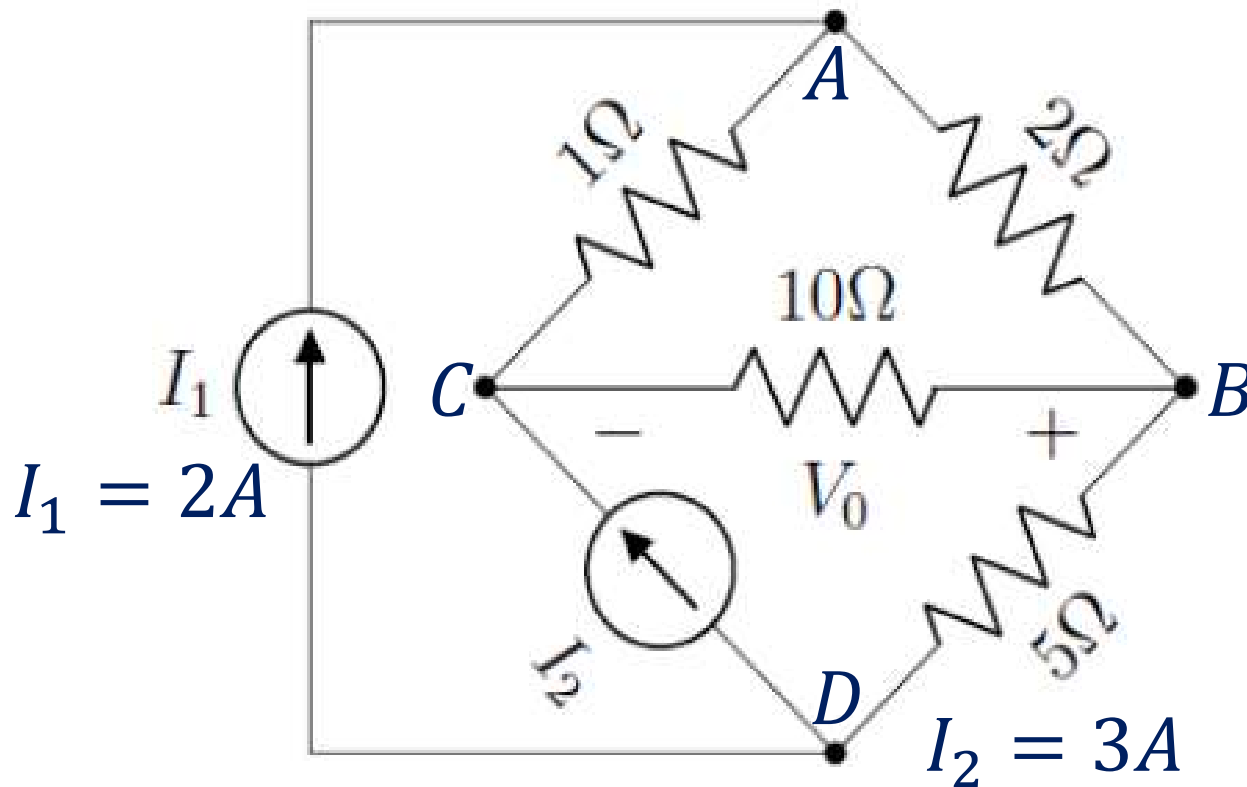
## Superposition Theorem: Only $I_2$ Active



$$V_{CB}(I_2) = V_{CAB}(I_2) = 3 \times \left( \frac{3 \times 10}{3 + 10} \right)$$

$$I_{CB}(I_2) = \frac{V_{CB}(I_2)}{10} = \frac{9}{13}$$

# Application of Superposition Theorem



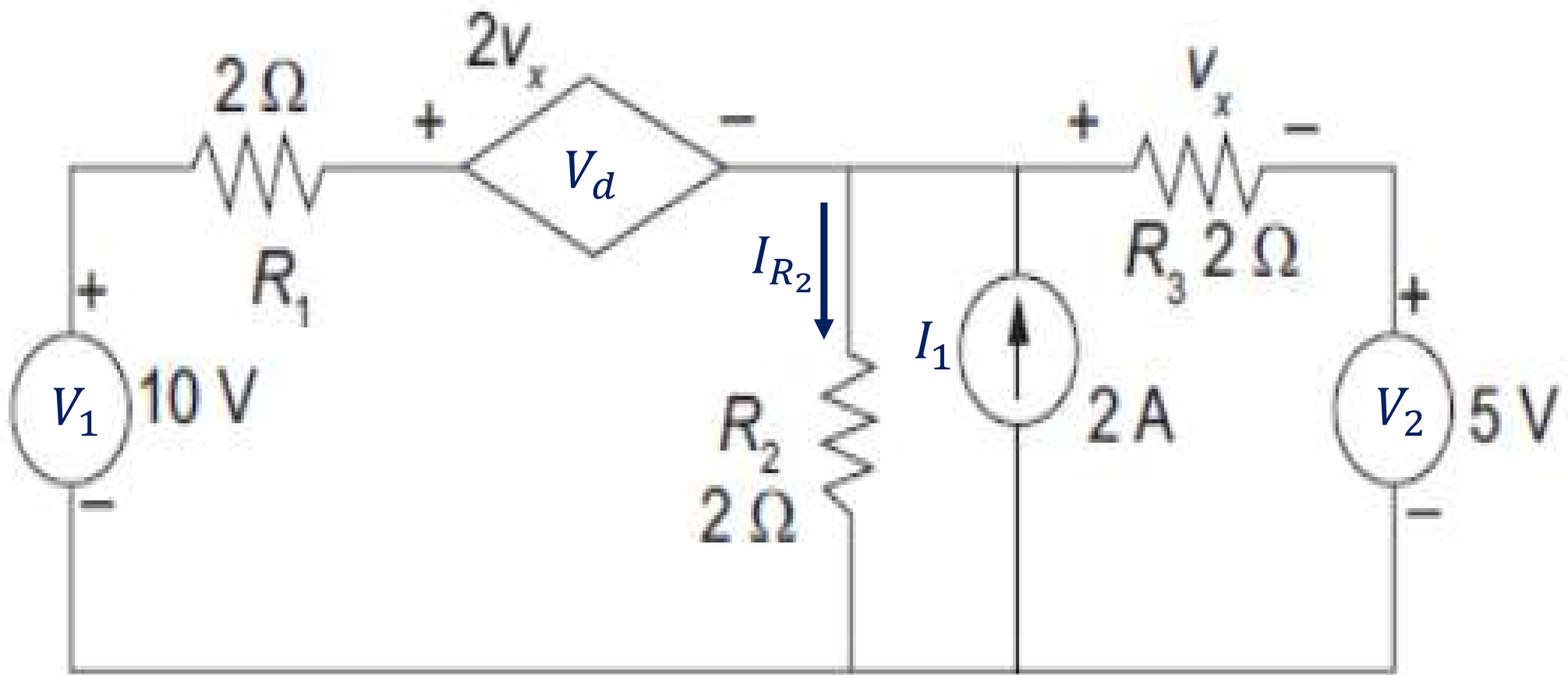
$$I_{CB} = I_{CB}(I_1) + I_{CB}(I_2)$$

$$I_{CB} = \frac{4}{13} + \frac{9}{13}$$

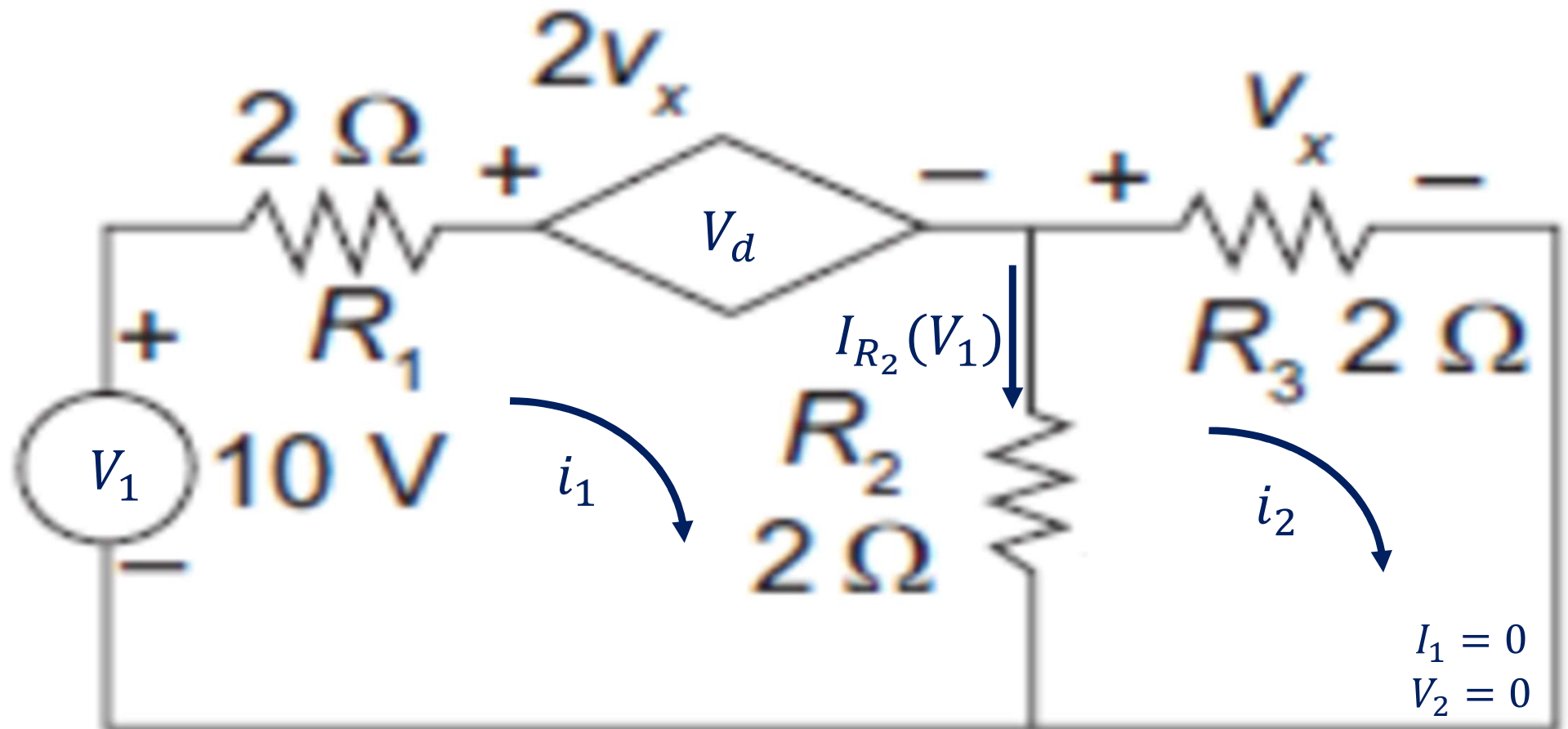
$$V_{BC} = -10 \times I_{CB} = -10V$$



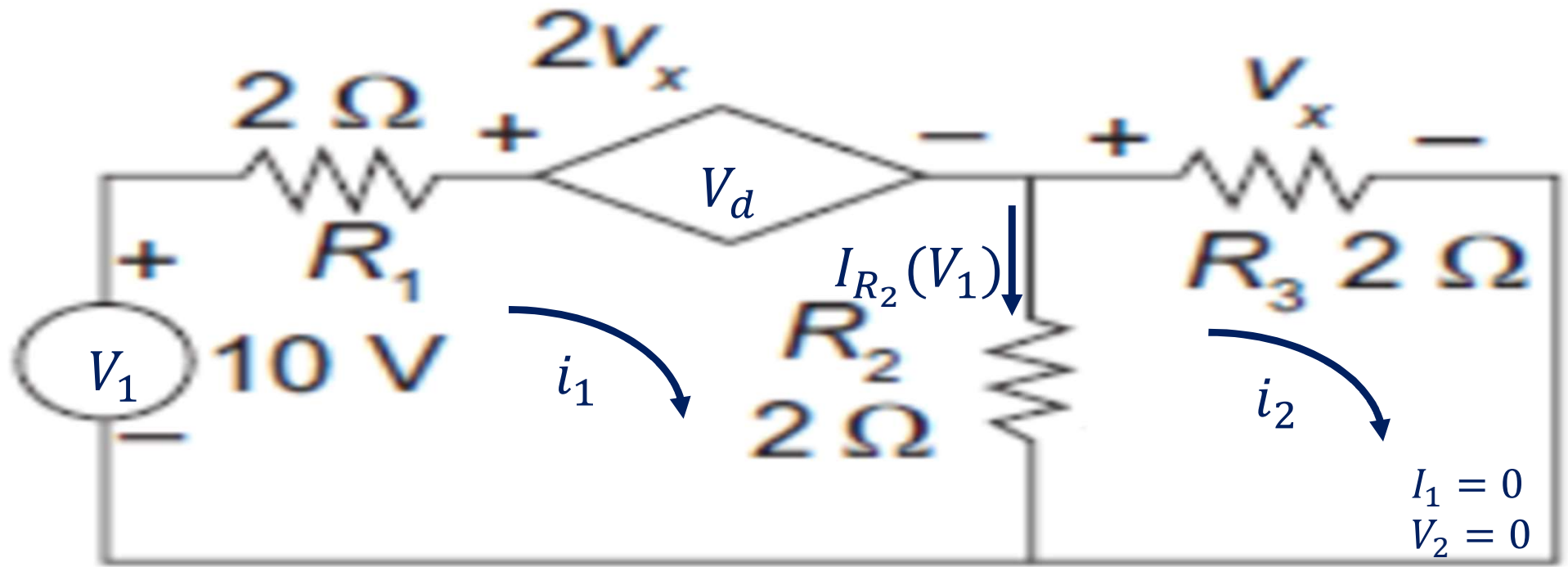
Solve for  $I_{R_2}$  Using Superposition Theorem



## Superposition Theorem: Only $V_1$ Active

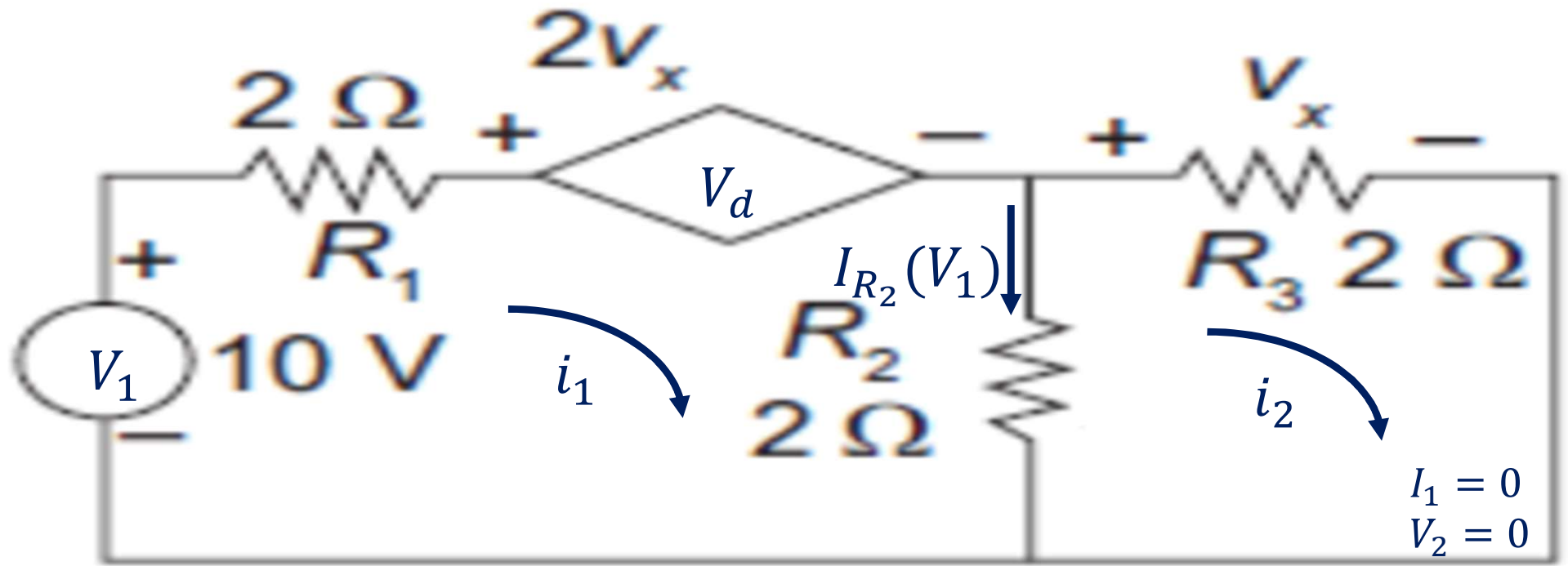


## Superposition Theorem: Only $V_1$ Active



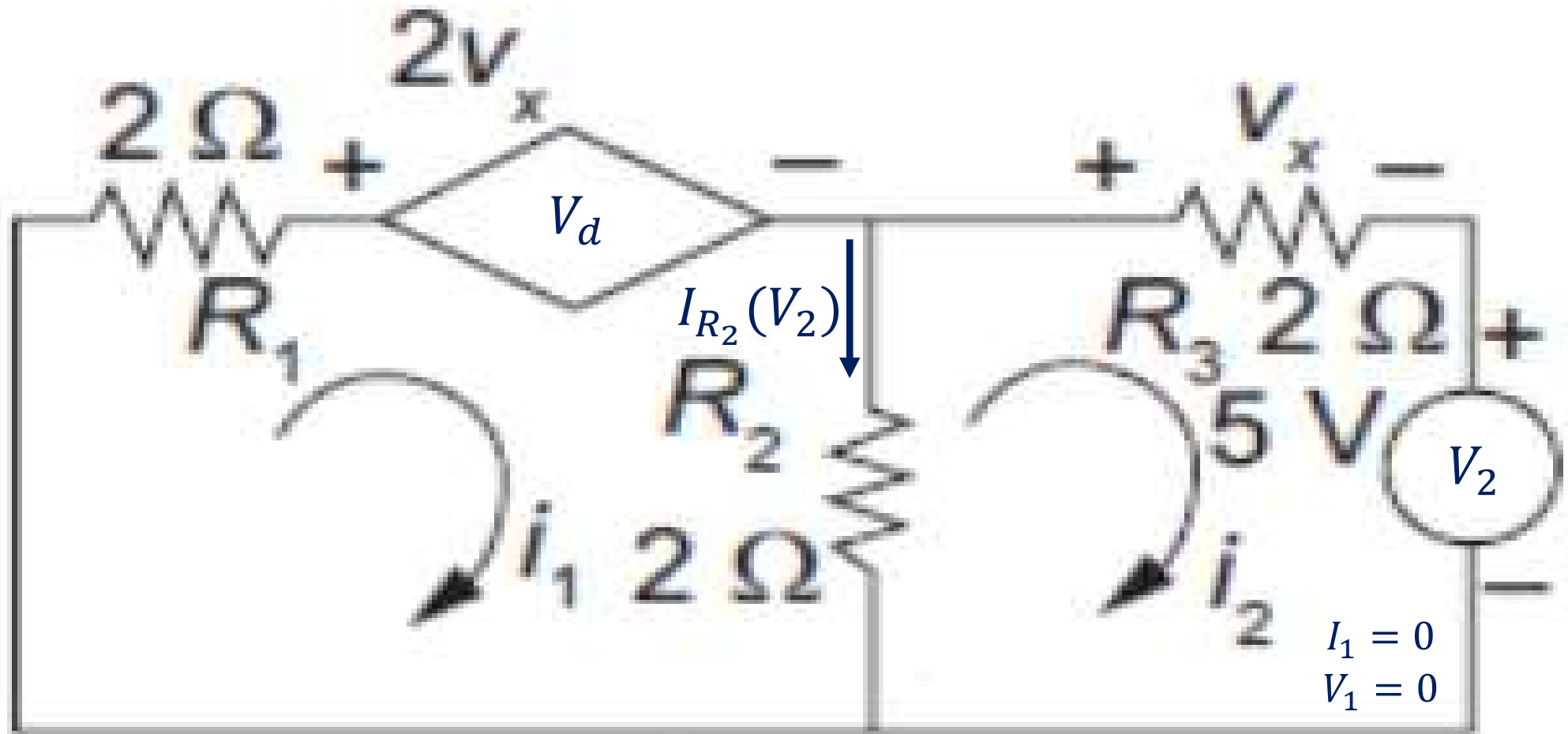
$$\begin{bmatrix} R_1 + R_2 & 2R_3 - R_2 \\ -R_2 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ 0 \end{bmatrix} \quad I_{R_2}(V_1) = i_1 - i_2$$

## Superposition Theorem: Only $V_1$ Active

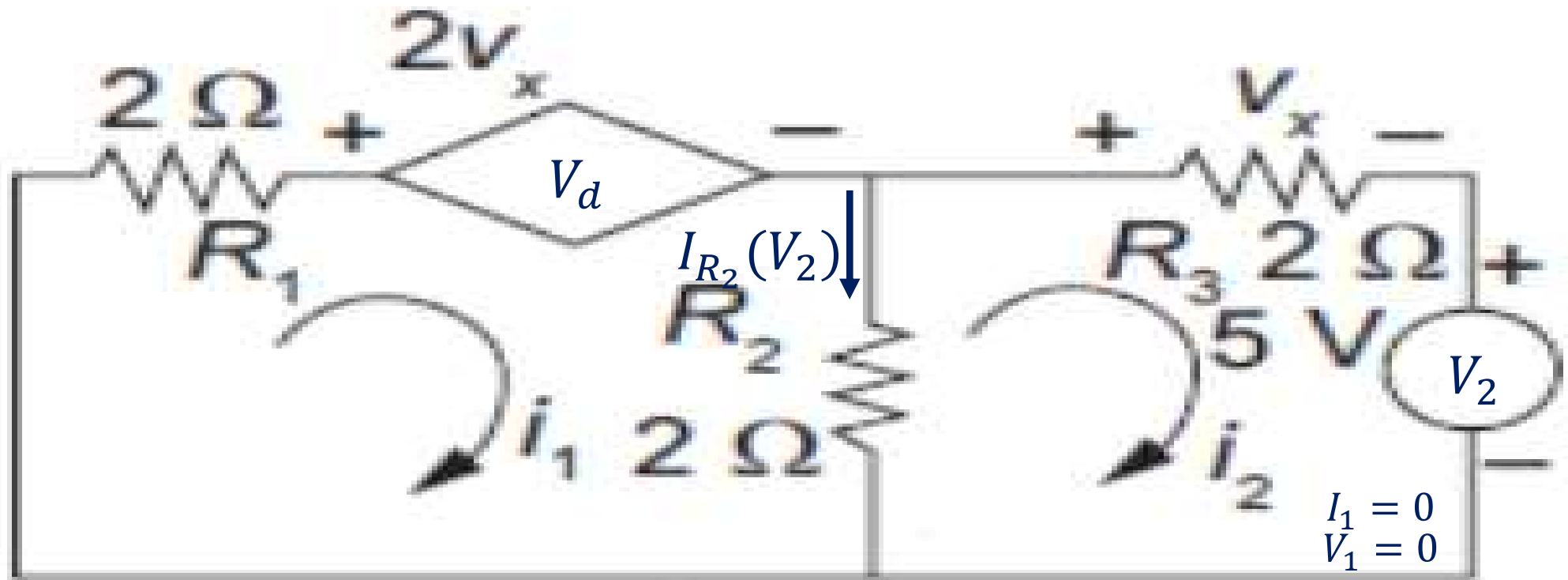


$$\begin{bmatrix} 4 & 2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad I_{R_2}(V_1) = i_1 - i_2 = 1$$

## Superposition Theorem: Only $V_2$ Active

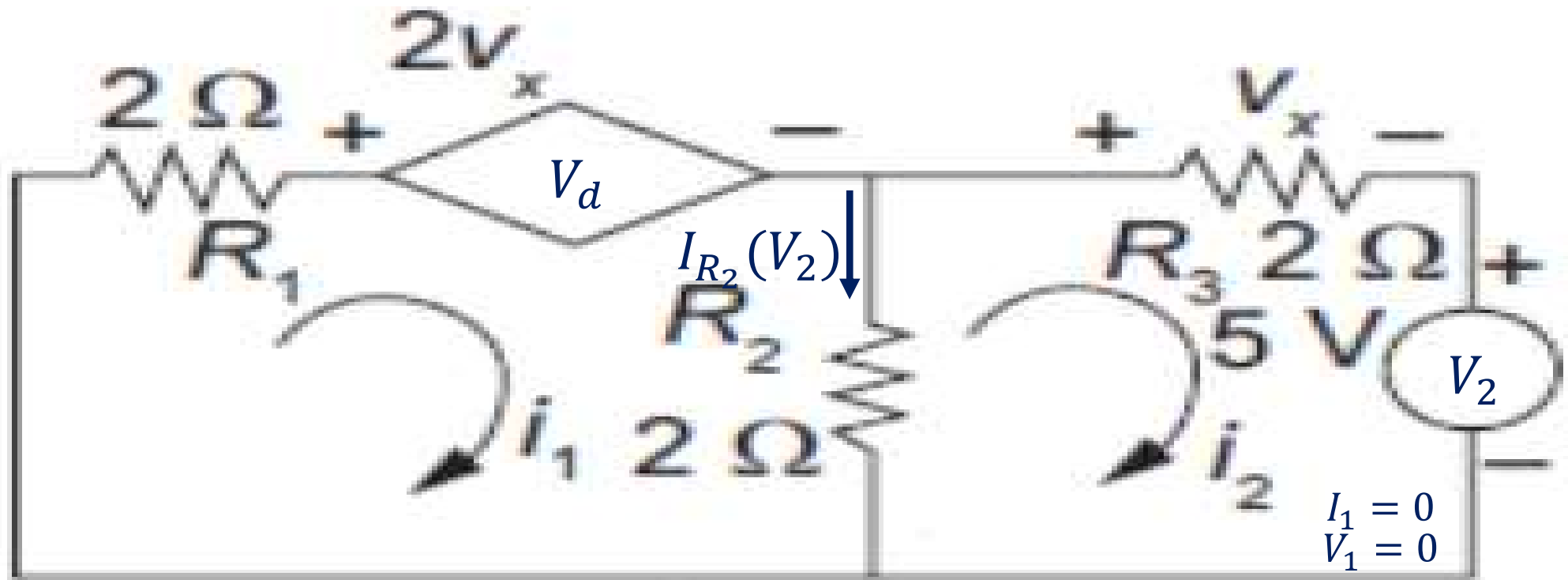


## Superposition Theorem: Only $V_2$ Active



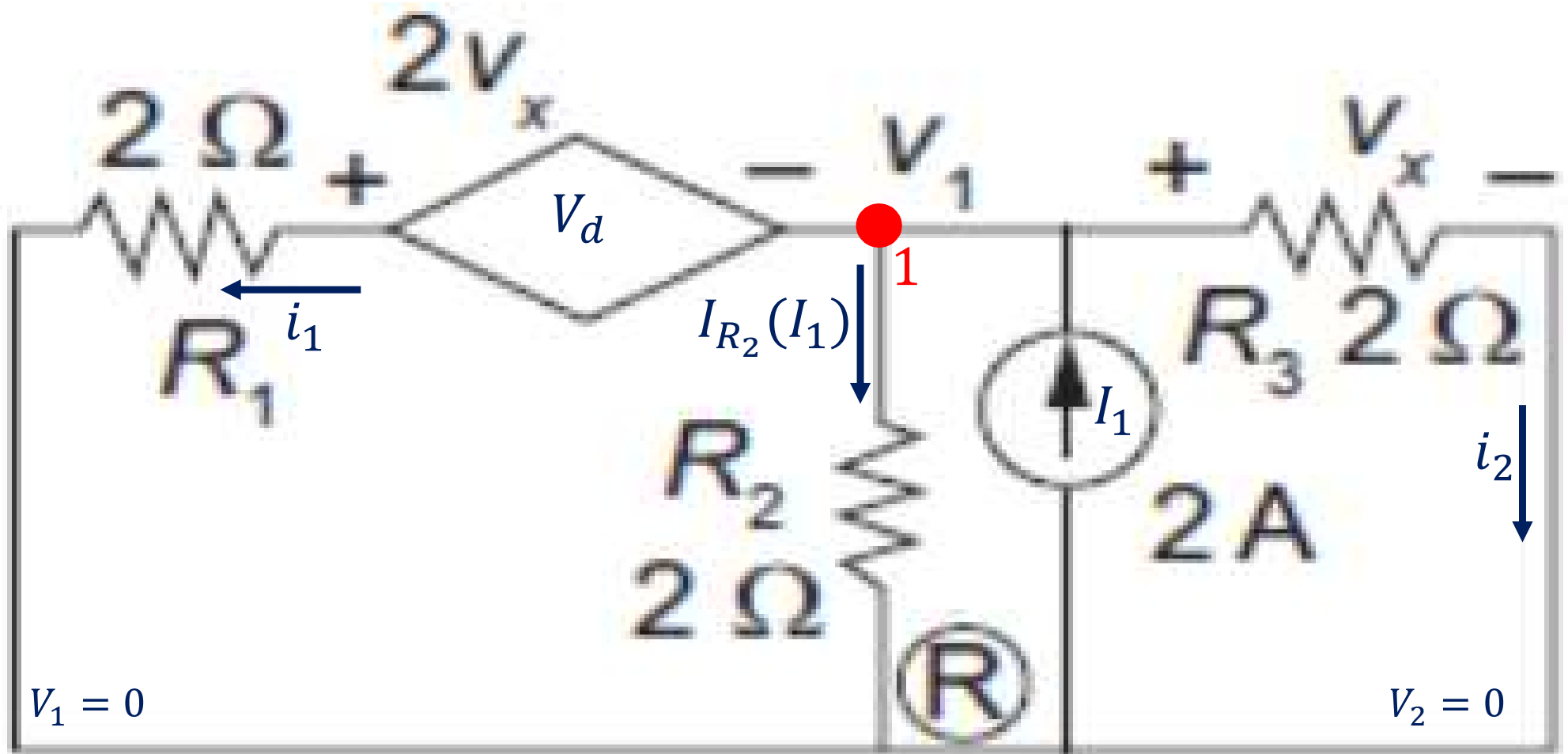
$$\begin{bmatrix} R_1 + R_2 & 2R_3 - R_2 \\ -R_2 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -V_2 \end{bmatrix} \quad I_{R_2}(V_2) = i_1 - i_2$$

## Superposition Theorem: Only $V_2$ Active



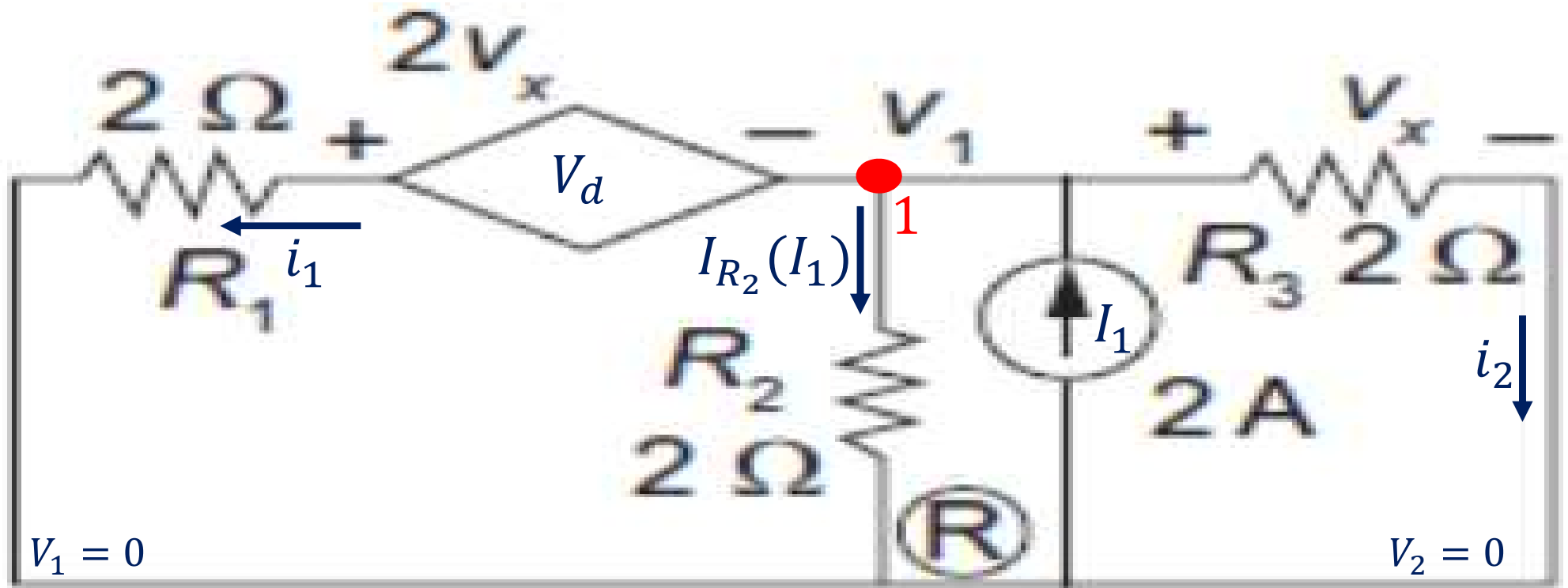
$$\begin{bmatrix} 4 & 2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \end{bmatrix} \Rightarrow \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -1 \end{bmatrix} \quad I_{R_2}(V_2) = i_1 - i_2 = 1.5$$

## Superposition Theorem: Only $I_1$ Active



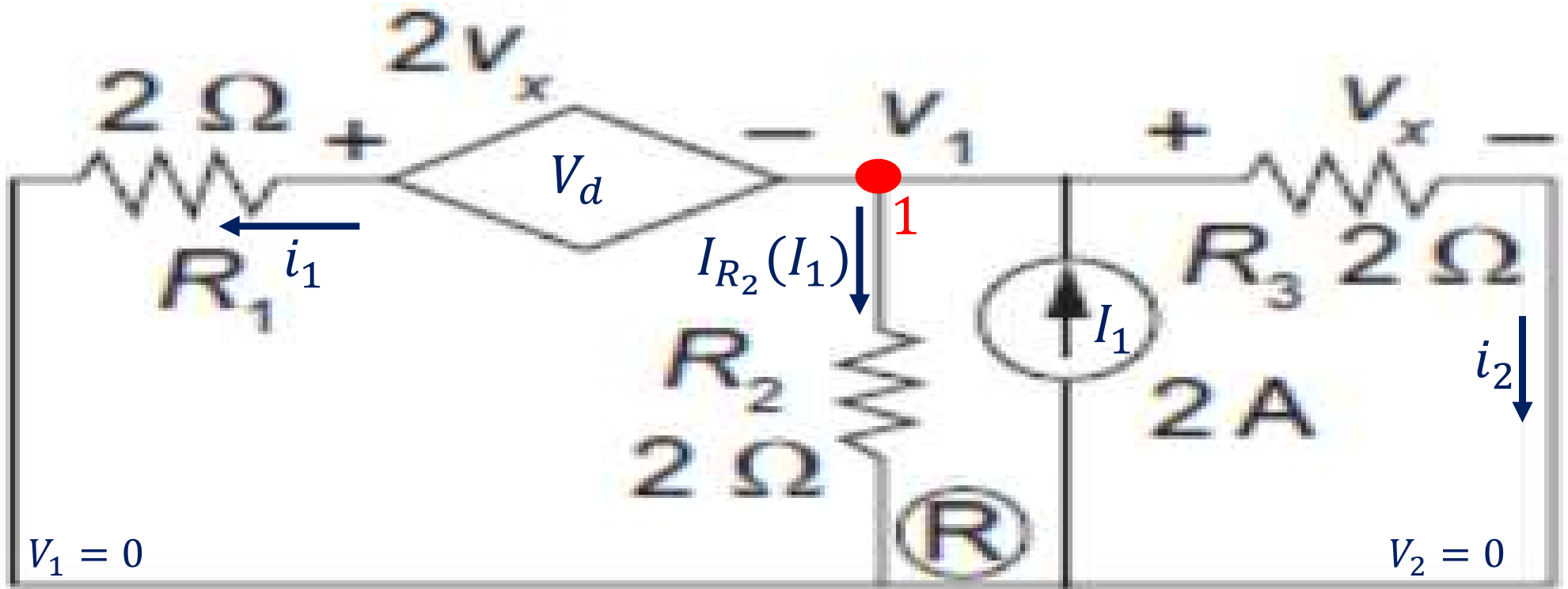


## Superposition Theorem: Only $I_1$ Active



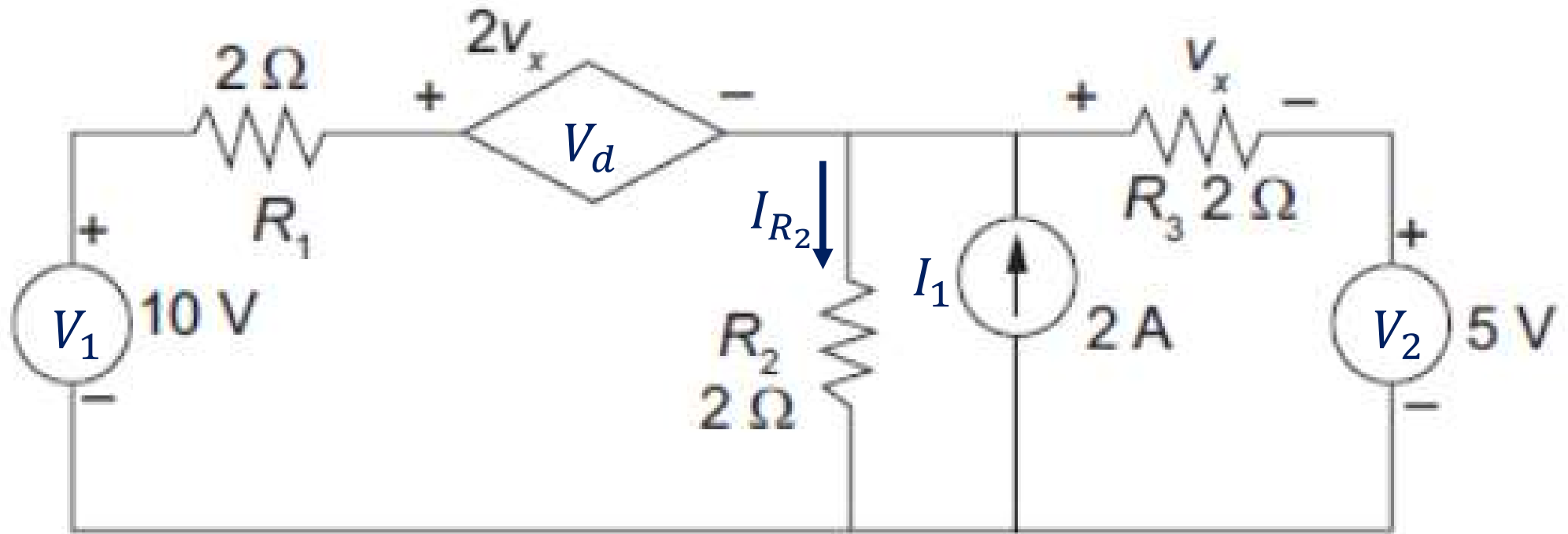
KCL at Node **1**:  $3v_1G_1 + G_2v_1 + G_3v_1 = I_1 \quad (v_1 = v_x)$

## Superposition Theorem: Only $I_1$ Active



$$3 \times 0.5v_1 + 0.5v_1 + 0.5v_1 = 2 \Rightarrow v_1 = 0.8 \Rightarrow I_{R_2}(I_1) = 0.4$$

## Application of Superposition Theorem



$$I_{R_2} = I_{R_2}(V_1) + I_{R_2}(V_2) + I_{R_2}(I_1) = 1 + 1.5 + 0.4 = 2.9$$

# Summary

- Numerical Problem Solving
  - Node Analysis
  - Mesh Analysis
- Application of Superposition Theorem
  - Circuits with only Independent Sources
  - Circuits with Dependent Sources



# Thank You